BEYOND THE BLACK BOX: DISENTANGLED AND CONTROLLABLE GENERATIVE MODELS FOR MEDICAL IMAGING

AXEL MARTIN ROMERO ANTOLIN

PhD in Artificial Intelligence
Facultat d'Informàtica de Barcelona (FIB)
Universitat Politècnica de Catalunya (UPC)

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SUPERVISORS:
Prof. Javier Béjar Alonso

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NOTATION

- **Scalars** are denoted by lowercase letters, e.g. $x \in \mathbb{R}$.
- **Vectors** are denoted by bold lowercase letters, e.g. $\mathbf{x} \in \mathbb{R}^n$.
- **Matrices** are denoted by bold uppercase letters, e.g. $\mathbf{X} \in \mathbb{R}^{m \times n}$.
- **Sets** are denoted by calligraphic uppercase letters, e.g. $\mathcal{X}$.
- $\mathbf{X}^\top$ denotes the transpose of a matrix $\mathbf{X}$.
- $\mathbf{I}$ denotes the identity matrix.
- $\mathbb{E}_{\mathbf{x} \sim p}[f(\mathbf{x})]$ denotes the expected value of a function $f(\mathbf{x})$ with respect to a probability distribution p over $\mathbf{x}$.
- $\mathrm{Var}_{\mathbf{x} \sim p}[f(\mathbf{x})]$ denotes the variance of a function $f(\mathbf{x})$ with respect to a probability distribution p over $\mathbf{x}$.
- $\nabla_\theta f(\theta)$ denotes the gradient of a function $f(\theta)$ with respect to the parameters θ.

1 INTRODUCTION

In recent years, generative artificial intelligence has fundamentally transformed the landscape of synthetic data generation. Early breakthroughs with Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs) demonstrated the potential of neural networks to learn and sample from complex data distributions. More recently, the emergence of denoising and score-based Diffusion Models and Flow Matching frameworks has set a new standard, achieving unprecedented fidelity and training stability when synthesizing highly realistic, high-dimensional data like images, which is the main focus of this thesis.
However, as these powerful models are increasingly used by the community and the industry, it has become clear that their raw generative capabilities are not sufficient for all applications. For high-risk domains such as healthcare, their inherently opaque, "black box" nature poses significant challenges for the deployment. This thesis aims to bridge the gap between raw generative power and clinical reliability. The primary focus of this research is to develop new methods to enhance the interpretability, disentanglement, and compositionality of state-of-the-art generative frameworks, such as Flow Matching and Diffusion models, for the generation of safe and reliable synthetic medical images. By establishing a deep theoretical understanding of the underlying principles of these models and the concepts of disentanglement and compositionality, this work identifies their current structural weaknesses and advantages and proposes solutions to overcome the complexities to be able to generate high-quality synthetic medical images to address the severe scarcity of annotated medical datasets.

The development and validation of robust medical AI systems require massive amounts of diverse, high-quality, and richly annotated data. However, the acquisition of such datasets is chronically hindered by strict privacy regulations, high annotation costs, and the inherent rarity of certain pathologies. While synthetic data generation offers a promising solution to this bottleneck, current standard generative models often lack the precise control necessary for clinical applications.
When a generative model operates as a "black box," users cannot reliably predict how specific anatomical structures or pathological features will be rendered. The widespread deployment of medical AI hinges fundamentally on the concept of human-AI trust. However,

as recent frameworks on AI trustworthiness emphasize, simply providing mathematical interpretability—trying to understand how the model works internally—is neither necessary nor sufficient for establishing this trust [44]. Instead, human trust in AI is built through interaction. Users must be able to run the model at will and probe its boundaries to observe predictable outcomes, generating empirical "behavior certificates" that prove the reliability of the model under specific constraints.
Therefore, to be truly safe and useful in a medical context, synthetic generation must prioritize this interactive capability through disentanglement. It must allow a clinician or researcher to alter a single semantic feature, such as the size, severity, or location of a tumor, and immediately observe the response of the model without the surrounding healthy anatomy becoming distorted. The motivation for this research stems from the critical need to facilitate this interactive trust. By developing frameworks where different semantic components can be independently manipulated and recombined, this thesis strives to transition generative models from static "black boxes" into highly interactive, trustworthy tools that users can rigorously test, understand through behavior, and safely use to simulate rare clinical scenarios.

To achieve the overarching goal of developing highly controllable and disentangled generative models for medical imaging, this research is driven by the following specific objectives:

- **Theoretical Grounding**: Acquire an in-depth theoretical understanding of diffusion and flow-based generative models, with a specific focus on their probabilistic foundations, interpolation strategies, and current structural limitations.
- **Comprehensive Literature Review**: Conduct a thorough analysis of existing literature surrounding control mechanisms and compositional frameworks in generative modeling, highlighting gaps in their application to the medical domain.
- **Baseline Benchmarking**: Reproduce and systematically benchmark baseline models, including Denoising Diffusion Probabilistic Models (DDPMs) and Flow Matching models, utilizing public and synthetic medical datasets to quantitatively evaluate current limitations in controllability and fidelity.
- **Design of Novel Control Strategies**: Propose and implement novel architectural modifications and training mechanisms, such as semantic conditioning, class guidance, and spatial control, to enforce strict user-defined constraints on the generative process.
- **Compositional Generation**: Develop compositional generation techniques that enable modular synthesis, ensuring that distinct

semantic components (e.g., pathology presence and anatomical structure) can be independently disentangled, manipulated, and seamlessly recombined.

- **Clinical Validation and Evaluation**: Apply the proposed models to real-world medical datasets, rigorously evaluating the methods through both quantitative metrics (e.g., FID, precision-recall, segmentation overlap) and qualitative assessments by clinical experts to ensure the utility, diversity, and realism of the generated data.
- **Dissemination and Open Science**: Contribute to the broader research community by disseminating findings through peer-reviewed publications and providing open-source implementations of the developed frameworks to ensure transparency and reproducibility.

Part I

FOUNDATIONS AND LITERATURE REVIEW

2 DEEP GENERATIVE MODELS

Let's start with the definition of our dataset $\mathcal{D}$, consisting of $N \geqslant 1$ data points, where the instances are assumed to be independently and identically distributed. Assume $\mathbf{x}$ represents the set of observed variables (images in our case), which is a random sample from an unknown underlying process, whose true distribution $p^*(\mathbf{x})$ is unknown. The idea is to approximate the real process with a model $p_\theta(\mathbf{x})$ with parameters θ. Learning has the aim to find a value of parameters θ such that the probability distribution of the model, $p_\theta(\mathbf{x})$, approximates the true distribution for any observed set of variables, i.e. $p_\theta(\mathbf{x}) \approx p^*(\mathbf{x})$.

Deep Generative Models (DGMs) are neural networks, which are a flexible way to parameterise distributions, that learn $p_\theta(\mathbf{x})$, which is commonly referred to as a generative model. Once the model is available, we can generate new samples using sampling methods and compute the likelihood of any given data point $\mathbf{x}'$ by evaluating $p_\theta(\mathbf{x}')$.

2.1 TRAINING OF DGM

The general idea is to learn the parameters θ by minimizing the discrepancy between real and predicted data distribution:

$$\theta^* = \arg\min_\theta \mathcal{D}(p_{data}, p_\theta).$$

Because p_{data} is unknown, we have to choose a discrepancy function that allows efficient estimation from samples from p_{data}.

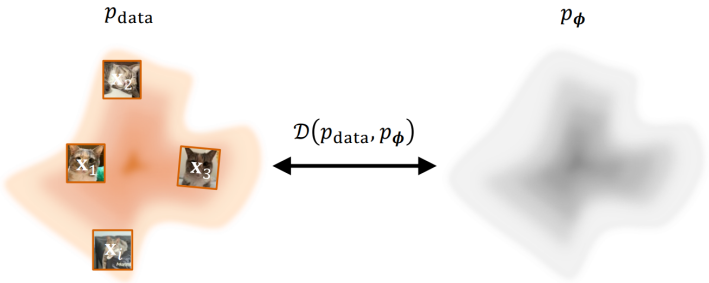


Figure 1: Visualization of the training of DGMs where we minimize a discrepancy measure between the real and unknown data distribution and the approximated one, using a set of i.i.d. samples.[29]

The most common choice is the Kullback-Leibler divergence defined as

$$\begin{aligned} \mathcal{D}_{KL}(p_{data}\|p_\theta) = \int p_{data}(\mathbf{x}) \log \frac{p_{data}(\mathbf{x})}{p_\theta(\mathbf{x})} d\mathbf{x} = \\ \mathbb{E}_{\mathbf{x}\sim p_{data}} \left[\log p_{data}(\mathbf{x}) - \log p_\theta(\mathbf{x})\right], \end{aligned} \tag{1}$$

where, taking into account only the part that depends on the parameter θ, minimizing the KL divergence is equivalent to performing MLE:

$$\min_\theta \mathcal{D}_{KL}(p_{data}\|p_\theta) \iff \max_\theta \mathbb{E}_{\mathbf{x}\sim p_{data}} \left[\log p_\theta(\mathbf{x})\right].$$

In practice, we do not have access to the data distribution, only to some i.i.d. samples $\{\mathbf{x}^{(i)}\}_{i=1}^N \sim p_{data}$, which are used to obtain the empirical estimation of the MSE

$$\mathcal{L}_{MLE}(\theta) = -\frac{1}{N}\sum_{i=1}^{N} \log p_\theta(\mathbf{x}^{(i)}).$$

There are some challenges in the use of neural networks to parameterize the probability distribution p_{data}. We can define a model that produces a real scalar $E_\theta(\mathbf{x}) \in \mathbb{R}$ for input $\mathbf{x}$. To interpret the output as a valid probability density, it has to satisfy the following conditions:

- **Non-Negativity**: $p_\theta(\mathbf{x}) \geqslant 0$ for all $\mathbf{x}$ in the domain. The standard positive function used to guarantee non-negativity is the exponential function, which produces an unnormalized density $\tilde{p}_\theta(\mathbf{x}) = \exp(E_\theta(\mathbf{x}))$.
- **Normalization**: The integral over the entire domain must equal one. To create a valid probability density, we must divide it by its integral over the entire space

  $$p_\theta(\mathbf{x}) = \frac{\tilde{p}_\theta(\mathbf{x})}{Z(\theta)},$$

  where the denominator is the partition function defined as

  $$Z(\theta) = \int \tilde{p}_\theta(\mathbf{x}')d\mathbf{x}'.$$

The partition function introduces a computational challenge in the learning of probabilistic models, especially for high-dimensional problems. This intractability is a central problem that motivates the development of many different families of deep generative models, which will be explained in the following chapters.

3

VARIATIONAL AUTOENCODERS

The core initial idea for the understanding of the Variational Autoencoders (VAEs) is the concept of **Autoencoders** (AEs). This are a type of model that contains two elements, the encoder and the decoder. The encoder is a function that maps the input data $\mathbf{x} \in \mathbb{R}^d$ to a low-dimensional representation $\mathbf{z} \in \mathbb{R}^k$, which is usually much more smaller than the input dimension, $k << d$. This representations live in a space with a more compact and interpretable structure, called the latent space. Then, the decoder is a function that maps the latent representation $\mathbf{z}$ back to the input space, trying to reconstruct the original data point $\mathbf{x}$.
Once we have an autoencoder model trained, a natural question is how to use it as a generative model. One idea is to randomly sample a point $\mathbf{z}$ from the latent space and then pass it through the decoder to get a synthetic data point $\mathbf{x}'$. This method will not produce realistic samples because the latent space is disorganized and irregular. Another option is to encode and image and sample neighboring points in the latent space that could generate similar samples, but the reality is that the output are not meaningful samples. This shows how poorly structured the latent space is. The aim of VAEs is to transform the latent space of an autoencoder into a well-structured space that follows a prior distribution, which allows us to sample from it and generate realistic samples.

It is also important to introduce the concept of latent variable models. This type of variables are part of the model but are not observed in the data, usually denoted as $\mathbf{z}$, which are the hidden factors of variation. The goal of latent variable models is to learn the joint distribution $p_\theta(\mathbf{x}, \mathbf{z})$ of the observed data $\mathbf{x}$ and the latent variables $\mathbf{z}$. The marginal likelihood, model evidence or marginal distribution over the observed data is given by:

$$p_\theta(\mathbf{x}) = \int p_\theta(\mathbf{x}, \mathbf{z}) d\mathbf{z}. \tag{2}$$

When we use neural networks to parameterize the latent variable models we call them **Deep Latent Variable Models** (DLVMs). The simple and most common DLVM is given by the following factorization:

$$p_\theta(\mathbf{x}, \mathbf{z}) = p(\mathbf{z}) p_\theta(\mathbf{x}|\mathbf{z}), \tag{3}$$

where $p(\mathbf{z})$ is the *prior distribution* over the latent variables and $p_\theta(\mathbf{x}|\mathbf{z})$ is the likelihood of the data given the latent variables, also called the

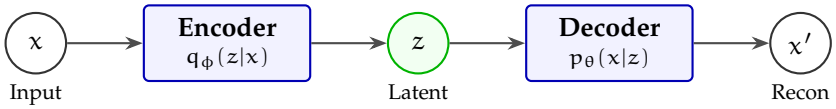


Figure 2: Simplified Variational Autoencoder (VAE) architecture mapping an input x to a latent representation z, and decoding it to a reconstruction x'.

generator distribution. The joint distribution $p_\theta(\mathbf{x}, \mathbf{z})$ is related through the Bayes' rule to the following posterior distribution:

$$p_\theta(\mathbf{z}|\mathbf{x}) = \frac{p_\theta(\mathbf{x}|\mathbf{z})p(\mathbf{z})}{p_\theta(\mathbf{x})}, \tag{4}$$

where we have an intractability problem due to the marginal likelihood $p_\theta(\mathbf{x})$ which is given by an integral over the latent variables (Equation **??**). To overcome this problem we can use variational inference (VI) which is a technique to approximate the posterior distribution $p_\theta(\mathbf{z}|\mathbf{x})$ with a simpler distribution $q_\phi(\mathbf{z}|\mathbf{x})$ that is parameterized by ϕ.

This is the main idea behind **Variational Autoencoders** (VAEs) [24] which are a type of DLVMs that use VI to learn the latent variable model. It defines a parametric inference model $q_\phi(\mathbf{z}|\mathbf{x})$, also called an *encoder*, where ϕ are the parameters. The idea is to optimize the parameters such that:

$$q_\phi(\mathbf{z}|\mathbf{x}) \approx p_\theta(\mathbf{z}|\mathbf{x}). \tag{5}$$

This VAE formulation allows us to transform an autoencoder model with an unstructured latent space into a generative model with a latent space that follows a prior distribution, typically a Gaussian distribution $p(\mathbf{z}) = \mathcal{N}(\mathbf{z}; \mathbf{0}, \mathbf{I})$, which is easy to sample from.

3.1 TRAINING OF VAES

For the training of the VAEs we can not directly optimize the marginal likelihood $p_\theta(\mathbf{x})$ due to the intractability problem, but we can optimize a lower bound of the marginal likelihood called the **Evidence Lower Bound** (ELBO). Let's derive the log-likelihood of the data

point $\mathbf{x}$ and see how we can get the ELBO using the Jensen's inequality [23]:

$$\begin{aligned}
\log p_\theta(\mathbf{x}) &= \log \int p_\theta(\mathbf{x}, \mathbf{z}) d\mathbf{z} && (6)\\
&= \log \int q_\phi(\mathbf{z}|\mathbf{x}) \frac{p_\theta(\mathbf{x}, \mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})} d\mathbf{z} && (7)\\
&= \log \mathbb{E}_{\mathbf{z}\sim q_\phi(\mathbf{z}|\mathbf{x})} \left[\frac{p_\theta(\mathbf{x}, \mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})} \right] && (8)\\
&\geqslant \mathbb{E}_{\mathbf{z}\sim q_\phi(\mathbf{z}|\mathbf{x})} \left[\log \frac{p_\theta(\mathbf{x}, \mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})} \right] && (9)\\
&= \mathbb{E}_{\mathbf{z}\sim q_\phi(\mathbf{z}|\mathbf{x})} \left[\log \frac{p_\theta(\mathbf{x}|\mathbf{z}) p(\mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})} \right] && (10)\\
&= \mathbb{E}_{\mathbf{z}\sim q_\phi(\mathbf{z}|\mathbf{x})} \left[\log \frac{p_\theta(\mathbf{x}|\mathbf{z})}{\frac{q_\phi(\mathbf{z}|\mathbf{x})}{p(\mathbf{z})}} \right] && (11)\\
&= \mathbb{E}_{\mathbf{z}\sim q_\phi(\mathbf{z}|\mathbf{x})} \left[\log p_\theta(\mathbf{x}|\mathbf{z}) - \log \frac{q_\phi(\mathbf{z}|\mathbf{x})}{p(\mathbf{z})} \right] && (12)\\
&= \underbrace{\mathbb{E}_{\mathbf{z}\sim q_\phi(\mathbf{z}|\mathbf{x})} \left[\log p_\theta(\mathbf{x}|\mathbf{z}) \right]}_{\text{Reconstruction}} - \underbrace{\mathbb{E}_{\mathbf{z}\sim q_\phi(\mathbf{z}|\mathbf{x})} \left[\log \frac{q_\phi(\mathbf{z}|\mathbf{x})}{p(\mathbf{z})} \right]}_{\text{KL Divergence}} && (13)
\end{aligned}$$

We can see that for any data point $\mathbf{x}$, the log-likelihood satisfies:

$$\begin{aligned}
\log p_\theta(\mathbf{x}) &\geqslant \mathcal{L}_{\mathrm{ELBO}}(\theta, \phi; \mathbf{x})\\
&= \mathbb{E}_{\mathbf{z}\sim q_\phi(\mathbf{z}|\mathbf{x})} \left[\log p_\theta(\mathbf{x}|\mathbf{z}) \right] - \mathcal{D}_{\mathrm{KL}}(q_\phi(\mathbf{z}|\mathbf{x}) || p(\mathbf{z})).
\end{aligned} \tag{14}$$

If we maximize the previous loss with respect to the parameters θ and ϕ, we are optimizing:

- **Reconstruction**: the first term encourage to maximize the likelihood $p_\theta(\mathbf{x}|\mathbf{z})$, which corresponds to the reconstruction of the data point $\mathbf{x}$ from the latent variables $\mathbf{z}$. Optimizing this term alone risks memorizing the training data, so we need extra regularization.
- **Latent KL Regularization**: the second term encourages the encoder distribution $q_\phi(\mathbf{z}|\mathbf{x})$ to be close to the prior distribution $p(\mathbf{z})$, which is easy to sample from.

Once we have the definition of the training objective, we can use stochastic gradient descent (SGD) to perform joint optimization w.r.t. all parameters θ and ϕ. However, we have a problem with the first term of the ELBO because we can not backpropagate through the sampling process of $\mathbf{z} \sim q_\phi(\mathbf{z}|\mathbf{x})$, as we can see in the left part of Figure **??**. To solve this problem we can use the **reparameterization trick** which consists in expressing the sampling process as a deterministic function of the parameters and some noise. For example, if

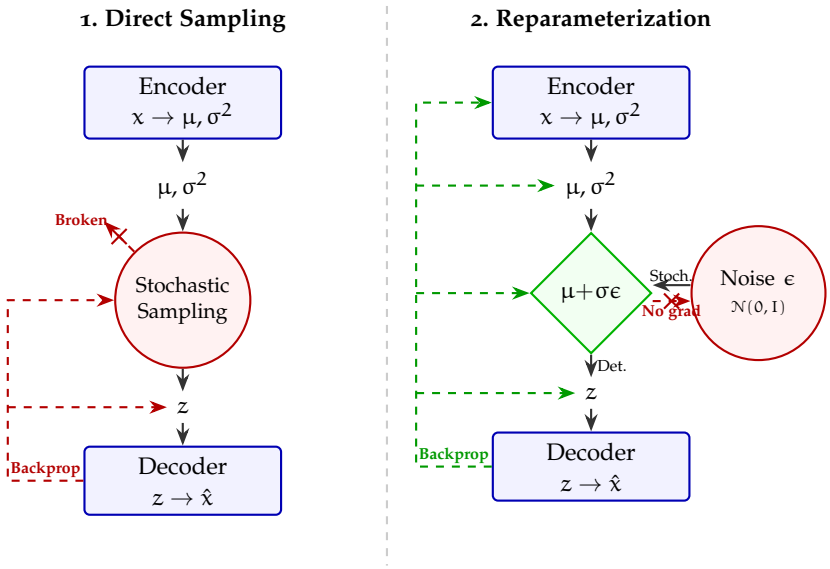


Figure 3: Reparameterization Trick in VAEs

we assume that $q_\phi(\mathbf{z}|\mathbf{x})$ is a Gaussian distribution with mean μ and variance σ^2, we can sample from it as follows:

$$\mathbf{z} = \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I). \tag{15}$$

This way, we can backpropagate through the deterministic function and optimize the parameters ϕ of the encoder. This is shown in the right part of Figure **??**.

3.2 GAUSSIAN VAES

The common choice for both the encoder and the decoder distributions in VAEs is the Gaussian distribution. This is because it allows us to use the reparameterization trick and also because it has a closed-form expression for the KL divergence.
The **encoder** $q_\phi(\mathbf{z}|\mathbf{x})$ is modelled as a Gaussian distribution defined as:

$$q_\phi(\mathbf{z}|\mathbf{x}) = \mathcal{N}\left(\mathbf{z}; \mu_\phi(\mathbf{x}), \text{diag}(\sigma^2_\phi(\mathbf{x}))\right), \tag{16}$$

where $\mu_\phi(\mathbf{x})$ and $\sigma^2_\phi(\mathbf{x})$ are the mean and variance of the Gaussian distribution, which are parameterized by a neural network with parameters ϕ.
The **decoder** $p_\theta(\mathbf{x}|\mathbf{z})$ is also modelled as a Gaussian distribution defined as:

$$p_\theta(\mathbf{x}|\mathbf{z}) = \mathcal{N}\left(\mathbf{x}; \mu_\theta(\mathbf{z}), \sigma^2\mathbf{I}\right), \tag{17}$$

where $\mu_\theta(\mathbf{z})$ is the mean of the Gaussian distribution, which is parameterized by a neural network with parameters θ, and $\sigma^2\mathbf{I}$ is the

covariance matrix, which is usually fixed to be a diagonal matrix with constant variance.

Under this Gaussian assumption, we can obtain a closed-form expression for the ELBO loss function. The reconstruction term can be expressed as:

$$\mathbb{E}_{\mathbf{z}\sim q_\phi(\mathbf{z}|\mathbf{x})}\left[\log p_\theta(\mathbf{x}|\mathbf{z})\right] = -\frac{1}{2\sigma^2}\mathbb{E}_{\mathbf{z}\sim q_\phi(\mathbf{z}|\mathbf{x})}\left[\|\mathbf{x}-\mu_\theta(\mathbf{z})\|^2\right] + C, \quad (18)$$

where C is a constant that does not depend on the parameters. The KL divergence term can be expressed as:

$$\mathcal{D}_{KL}(q_\phi(\mathbf{z}|\mathbf{x})\|p(\mathbf{z})) = \frac{1}{2}\sum_{i=1}^{k}\left(\sigma^2_{\phi,i}(\mathbf{x}) + \mu^2_{\phi,i}(\mathbf{x}) - 1 - \log\sigma^2_{\phi,i}(\mathbf{x})\right), \quad (19)$$

where k is the dimension of the latent space. Therefore, the ELBO loss function can be expressed as:

$$\mathcal{L}_{ELBO}(\theta,\phi;\mathbf{x}) = -\frac{1}{2\sigma^2}\mathbb{E}_{\mathbf{z}\sim q_\phi(\mathbf{z}|\mathbf{x})}\left[\|\mathbf{x}-\mu_\theta(\mathbf{z})\|^2\right] \quad (20)$$

$$-\frac{1}{2}\sum_{i=1}^{k}\left(\sigma^2_{\phi,i}(\mathbf{x}) + \mu^2_{\phi,i}(\mathbf{x}) - 1 - \log\sigma^2_{\phi,i}(\mathbf{x})\right) + C. \quad (21)$$

Despite their mathematical elegance and continuous latent spaces, standard Gaussian Variational Autoencoders notoriously suffer from generating blurry images. This limitation fundamentally stems from the VAE objective function and its underlying statistical assumptions. The problem can be attributed to four primary factors:

- **MSE Loss**: In a standard Gaussian VAE with a fixed variance $\sigma^2\mathbf{I}$, maximizing the likelihood $p_\theta(\mathbf{x}|\mathbf{z})$ is mathematically equivalent to minimizing the $\mathcal{L}_2$ distance between the input $\mathbf{x}$ and the reconstruction $\hat{\mathbf{x}}$:

  $$\log p_\theta(\mathbf{x}|\mathbf{z}) \propto -\|\mathbf{x}-\mu_\theta(\mathbf{z})\|_2^2 \quad (22)$$

  The $\mathcal{L}_2$ loss heavily penalizes large pixel-wise deviations. When the model is uncertain about the exact location of high-frequency details (such as sharp edges), it outputs the statistical mean of all plausible variations to minimize the overall penalty. This phenomenon, known as *mode averaging*, results in safe, blurry predictions rather than rendering sharp, distinct features.

- **Information Bottleneck**: The ELBO includes a regularization term that minimizes the KL divergence between the approximate posterior and the prior:

  $$\mathcal{D}_{KL}(q_\phi(\mathbf{z}|\mathbf{x}) \parallel p(\mathbf{z})) \quad (23)$$

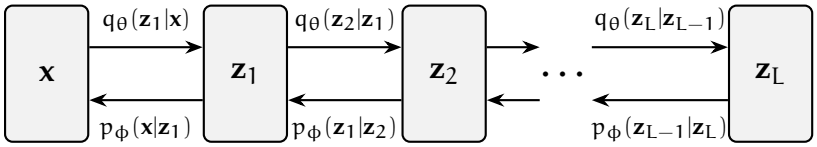


Figure 4: Hierarchical VAE architecture with L layers of latent variables. Each layer captures different levels of abstraction, allowing for richer representations and improved generation quality.

This term acts as an information bottleneck, forcing the encoded latent representations to closely match the simple prior, typically $\mathcal{N}(\mathbf{0}, \mathbf{I})$. Consequently, the encoder discards complex, high-frequency spatial features that are difficult to compress, prioritizing global structures and low-frequency components instead.

- **Posterior Collapse**: Furthermore, consistent with findings in [2] and [47], stochastic optimization using the unmodified lower bound objective frequently gets stuck in an undesirable stable equilibrium. At the onset of training, the reconstruction likelihood term $\log p_\theta(\mathbf{x}|\mathbf{z})$ is relatively weak. Consequently, an initially attractive state for the network is to aggressively minimize the latent cost, driving the approximate posterior to match the prior, $q_\phi(\mathbf{z}|\mathbf{x}) \approx p(\mathbf{z})$. This results in a stable equilibrium—often termed *posterior collapse*—from which the model struggles to escape, yielding uninformative latent representations and contributing to poor generation quality. To mitigate this, a common solution proposed in [2, 47] is to implement an optimization schedule (KL annealing) where the weight of the latent cost $\mathcal{D}_{KL}(q_\phi(\mathbf{z}|\mathbf{x}) \parallel p(\mathbf{z}))$ is slowly annealed from 0 to 1 over multiple epochs.

3.3 DENOISING DIFFUSION PROBABILISTIC MODELS

One important technique developed to enhance the performance of VAEs and overcome the main limitations was **Hierarchical Variational Autoencoders** (HVAEs) [48] which are a type of VAE that uses a hierarchical structure in the latent space. This hierarchical structure allows the model to capture more complex and high-frequency details in the data, which can help to reduce the blurriness of the generated images. A visualization of the architecture of HVAEs is shown in Figure **??**. In this architecture, we have L layers of latent variables $\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_L$, where each layer captures different levels of abstraction.
In this setup, we can define the following factorization of the joint distribution:

$$p_\theta(\mathbf{x}, \mathbf{z}_{1:L}) = p_\theta(\mathbf{x}|\mathbf{z}_1) \prod_{i=2}^{L} p_\theta(\mathbf{z}_{i-1}|\mathbf{z}_i) p(\mathbf{z}_L), \tag{24}$$

and the marginal distribution of the data point $\mathbf{x}$ can be expressed as:

$$p_\theta(\mathbf{x}) = \int p_\theta(\mathbf{x}, \mathbf{z}_{1:L}) d\mathbf{z}_{1:L}, \tag{25}$$

where the generation is performed progressively from the latent variable $\mathbf{z}_L$, which follows the prio distribution $p(\mathbf{z}_L)$, to the data point $\mathbf{x}$, which is generated from the latent variable $\mathbf{z}_1$. The inference model can be defined as:

$$q_\phi(\mathbf{z}_{1:L}|\mathbf{x}) = q_\phi(\mathbf{z}_1|\mathbf{x}) \prod_{i=2}^{L} q_\phi(\mathbf{z}_i|\mathbf{z}_{i-1}), \tag{26}$$

where the inference is performed progressively from the data point $\mathbf{x}$ to the latent variable $\mathbf{z}_L$.
The ELBO loss is very similar to the one of the standard VAE, but we have to consider all the layers of latent variables:

$$\mathcal{L}_{ELBO}(\theta, \phi; \mathbf{x}) = \mathbb{E}_{\mathbf{z}_{1:L} \sim q_\phi(\mathbf{z}_{1:L}|\mathbf{x})} \left[\log \frac{p_\theta(\mathbf{x}, \mathbf{z}_{1:L})}{q_\phi(\mathbf{z}_{1:L}|\mathbf{x})} \right], \tag{27}$$

which can also be decomposed into the reconstruction term and the KL divergence term.

This method of deep, stacked layers revolutionized the field of generative modeling and is the basis for many state-of-the-art models. The training of HVAEs present several challenges because the encoder and decoder must be trained jointly and learning becomes unstable. A clever twist to solve the main problems and maintain the hierarchical structure is the **Denoising Diffusion Probabilistic Models** (DDPMs) [12, 46], the next step in generative modeling, that can be defined with the following stochastic processes.

3.3.1 *Forward Pass*

This is a fixed encoder (not learnable) that progressively adds noise to the data point $\mathbf{x}$ via a transition kernel $q(\mathbf{x}_t|\mathbf{x}_{t-1})$ until we get pure noise $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. The fixed Gaussian transition kernel is defined as:

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t \mathbf{I}), \tag{28}$$

where $\mathbf{x}_0 \sim p_{data}$, $\mathbf{x}_T \sim p_{prior} = \mathcal{N}(\mathbf{0}, \mathbf{I})$ and the sequence $\{\beta_t\}_{t=1}^T \in (0, 1)$ is the noise scheduler, which controls the amount of noise added at each step, as we can see in the following alternative formulation:

$$\mathbf{x}_t = \sqrt{\alpha_t}\mathbf{x}_{t-1} + \sqrt{1-\alpha_t}\epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \tag{29}$$

where $\alpha_t = 1 - \beta_t$. There are different options of noise schedulers that can be used, where the original DDPM paper [12] use a linear scheduler from $\beta_1 = 10^{-4}$ to $\beta_T = 0.02$. Following studies demonstrate that the use of more advanced schedulers can improve the performance of the model, such as the cosine scheduler, which is widely used in the literature [5, 39].

Another important formulation of the forward pass can be achieve by recursively applying the transition kernel (Equation **??**), where we achieve a closed-form expression for the distribution of noisy samples at step t given the original data:

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1 - \bar{\alpha}_t)\mathbf{I}), \tag{30}$$

where $\bar{\alpha}_t = \prod_{s=1}^{T} \alpha_s$. This formulation is very useful because it allows us to sample from the forward process at any time step t without having to recursively apply the transition kernel t times. We can also define the sampling of $\mathbf{x}_t$ as in Equation **??** as follows:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \tag{31}$$

By definition $0 < \beta_t < 1$, therefore $\alpha_t < 1$ and, if we have a lot of time steps ($T \to \infty$), then $\prod_{t=1}^{T} \alpha_t \to 0$, which implies that the distribution of the higher time step latent variable takes a simple form $p_T(\mathbf{x}_T) \to \mathcal{N}(\mathbf{x}_T; \mathbf{0}, \mathbf{I}) = p_{prior}$. This shows the need of a large number of time steps to ensure that the distribution of the noisy samples at the last time step is close to a simple distribution, which is important for the reverse pass. In the original DDPM paper [12], the authors use $T = 1000$ time steps, which is a common practice in the literature.

3.3.2 *Reverse Pass*

This is the decoder, where a neural network is trained to reverse the noising process by learning the parameterized distribution $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$. If we start from a pure noise sample, we can iteratively apply the reverse process to generate a data point $\mathbf{x}_0$ that resembles the training data, following a Markov chain. The training goal is to approximate the unknown reverse transition kernel $p(\mathbf{x}_{t-1}|\mathbf{x}_t)$ with a learnable parametric model $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$, using the KL divergence as the training objective:

$$\mathbb{E}_{\mathbf{x}_t \sim p_t}\left[\mathcal{D}_{KL}(p(\mathbf{x}_{t-1}|\mathbf{x}_t)||p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))\right]. \tag{32}$$

We can express the unknown reverse transition kernel $p(\mathbf{x}_{t-1}|\mathbf{x}_t)$ in terms of the forward process using Bayes' rule:

$$p(\mathbf{x}_{t-1}|\mathbf{x}_t) = \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1})p_{t-1}(\mathbf{x}_{t-1})}{p_t(\mathbf{x}_t)}, \tag{33}$$

where the marginal distribution $p_t(\mathbf{x}_t)$ can be expressed as:

$$p_t(\mathbf{x}_t) = \int q(\mathbf{x}_t|\mathbf{x}_0)p_{data}(\mathbf{x}_0)d\mathbf{x}_0. \tag{34}$$

Since p_{data} is unknown, we can not compute the marginal distribution $p_t(\mathbf{x}_t)$ thus the computation of the target distribution $p(\mathbf{x}_{t-1}|\mathbf{x}_t)$ is intractable. A trick to solve this problem is to condition the reverse transition kernel on the original data point $\mathbf{x}_0$. Now we can define the target distribution as follows:

$$p(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1})q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)}, \tag{35}$$

which allows us to have a tractable expression and to derive a tractable objective equivalent to the previous intractable KL divergence in Equation **??**. As a result, $p(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$ is a Gaussian distribution with mean and variance that can be computed in closed-form:

$$p(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}; \tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0, t), \tilde{\beta}_t\mathbf{I}), \tag{36}$$

where

$$\tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0, t) = \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}\mathbf{x}_0 + \frac{(1-\bar{\alpha}_{t-1})\sqrt{\alpha}_t}{1-\bar{\alpha}_t}\mathbf{x}_t, \quad \tilde{\beta}_t = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t. \tag{37}$$

Recall that the previous closed form expression dependes on the original data point $\mathbf{x}_0$, which is unknown. Because of this, we approximate the target distribution $p(\mathbf{x}_{t-1}|\mathbf{x}_t)$ with a learnable parametric model $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$, which is a Gaussian distribution with mean $\mu_\theta(\mathbf{x}_t, t)$ and variance $\Sigma_\theta(\mathbf{x}_t, t)\mathbf{I}$, where the variance is usually fixed and the mean is parameterized by a neural network with parameters θ. It can be defined as

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t), \sigma^2(t)\mathbf{I}). \tag{38}$$

Using the previous conditional derivation of the unknown reverse transition kernel, we can define a closed-form expression for the KL divergence training loss (Equation **??**) because minimizing the KL divergence between marginal distribution is mathematically identical to minimizing the KL divergence between specific conditional distributions. We can define the training loss as follows:

$$\mathcal{L}(\mathbf{x}_0; \theta) = \mathbb{E}_{t,\mathbf{x}_0}\left[\frac{1}{2\sigma^2(t)}||\mu_\theta(\mathbf{x}_t, t) - \tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0, t)||^2 + C\right]. \tag{39}$$

where C is a constant that does not depend on the parameters. This loss can be interpreted as a weighted $\mathcal{L}_2$ loss between the predicted mean $\mu_\theta(\mathbf{x}_t, t)$ and the true mean $\tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0, t)$ of the reverse transition kernel.

In practice, we do not use the loss of Equation **??** directly. If we combine the definition of the true mean $\tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0, t)$ in Equation **??** with Equation **??**, we can obtain two different parameterizations of the mean that can be used to derive two different but equivalent training losses that are easier to optimize.

- ϵ-Prediction. From Equation **??** we obtain $\mathbf{x}_0 = \frac{1}{\sqrt{\bar{\alpha}_t}}(\mathbf{x}_t - \sqrt{1-\bar{\alpha}_t}\epsilon)$ and we can substitute this expression in the definition of the true mean $\tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0, t)$ to obtain the following expression:

$$\tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0, t) = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon\right), \tag{40}$$

  which can also be applied to parameterized mean using now a neural network that predicts the noise $\epsilon_\theta(\mathbf{x}_t, t)$ instead of the mean $\mu_\theta(\mathbf{x}_t, t)$:

$$\mu_\theta(\mathbf{x}_t, t) = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(\mathbf{x}_t, t)\right). \tag{41}$$

  We can define a new loss function based on the noise prediction as follows:

$$\mathcal{L}(\mathbf{x}_0; \theta) = \mathbb{E}_{t,\mathbf{x}_0,\epsilon}\left[\frac{\beta_t^2}{2\sigma^2(t)\alpha_t(1-\bar{\alpha}_t)}\|\epsilon_\theta(\mathbf{x}_t, t) - \epsilon\|^2\right], \tag{42}$$

  where a simple version of this loss, where the weighting factor is removed, is commonly used in practice for the good performance and stability of the training process found in [12]. The final loss can be expressed as:

$$\mathcal{L}(\mathbf{x}_0; \theta) = \mathbb{E}_{t,\mathbf{x}_0,\epsilon}\left[\|\epsilon_\theta(\mathbf{x}_t, t) - \epsilon\|^2\right]. \tag{43}$$

- $\mathbf{x}_0$-Prediction: With the definition of the true mean $\tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0, t)$ (Equation **??**) we can obtain a new parameterization of the mean by using a neural network that predicts the original data point $\mathbf{x}_0$ instead of the mean $\mu_\theta(\mathbf{x}_t, t)$:

$$\mu_\theta(\mathbf{x}_t, t) = \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}\mathbf{x}_\theta(\mathbf{x}_t, t) + \frac{(1-\bar{\alpha}_{t-1})\sqrt{\alpha_t}}{1-\bar{\alpha}_t}\mathbf{x}_t, \tag{44}$$

  which, as in the previous case, can be used to derive a new loss function based on the original data point prediction as follows:

$$\mathcal{L}(\mathbf{x}_0; \theta) = \mathbb{E}_{t,\mathbf{x}_0,\epsilon}\left[\omega_t\|\mathbf{x}_\theta(\mathbf{x}_t, t) - \mathbf{x}_0\|^2\right]. \tag{45}$$

  where ω_t is a weighting factor that depends on the noise schedule defined as $\omega_t(\lambda_t) = \exp(\lambda_t)$ where $\lambda_t = \log(\bar{\alpha}_t/(1-\bar{\alpha}_t))$, which is called the signal-to-noise ratio (SNR) of the noisy sample $\mathbf{x}_t$ at time step t.

For the training of the model, we have to sample a data point $\mathbf{x}_0$ from the training data, a time step t from a uniform distribution over the time steps and a noise ϵ from a standard Gaussian distribution. Then we can compute the noisy sample $\mathbf{x}_t$ using the forward process and take a gradient descent step on the loss function defined in one of the two previous parameterizations. The training algorithm can be found in Algorithm **??**.

Algorithm 1 Training in DDPMs

repeat
- $\mathbf{x}_0 \sim p_{data}$
- $t \sim \text{Uniform}(\{1, \dots, T\})$
- $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
- Take gradient descent step on
- $\nabla_\theta \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta \left(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, t \right) \right\|^2$

until *converged*;

3.3.3 *Sampling*

Assume we have a trained model that predicts the noise, denoted as $\epsilon^*_\theta(\mathbf{x}_t, t)$. We can generate a new data point by starting from a pure noise sample $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and iteratively applying the reverse process $p^*_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ for $t = T, \dots, 1$, following the update rule defined as follows:

$$\mathbf{x}_{t-1} \leftarrow \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon^*_\theta(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}, \tag{46}$$

where $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ is a noise sample added to the reverse process to ensure stochasticity in the generation and σ_t is a hyperparameter that controls the amount of noise added at each step. The sampling algorithm can be found in Algorithm **??**. One of the main limitations of this sampling process is that it is very slow because we have to apply the reverse process T times, which can be computationally expensive. The early steps set the global structure of the generated image, while the later steps add finer details. Therefore, reducing the number of steps can lead to a significant decrease in generation quality.

Algorithm 2 Sampling in DDPMs

$\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
for $t = T, \dots, 1$ **do**
- $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $t > 1$, else $\mathbf{z} = \mathbf{0}$
- $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$

return $\mathbf{x}_0$

Summary of Key Equations: VAEs and DDPMs

This box summarizes the main mathematical equations of VAEs and DDPMs:

1. Training Objective of VAEs:

$$\mathcal{L}_{\mathrm{ELBO}}(\theta, \phi; \mathbf{x}) = \mathbb{E}_{\mathbf{z}\sim q_\phi(\mathbf{z}|\mathbf{x})}\left[\log p_\theta(\mathbf{x}|\mathbf{z})\right] - \mathcal{D}_{\mathrm{KL}}(q_\phi(\mathbf{z}|\mathbf{x})||p(\mathbf{z})). \tag{47}$$

2. Training Objective of Gaussian VAEs:

$$\mathcal{L}_{\mathrm{ELBO}}(\theta, \phi; \mathbf{x}) = -\frac{1}{2\sigma^2}\mathbb{E}_{\mathbf{z}\sim q_\phi(\mathbf{z}|\mathbf{x})}\left[||\mathbf{x} - \mu_\theta(\mathbf{z})||^2\right] \tag{48}$$

$$-\frac{1}{2}\sum_{i=1}^{k}\left(\sigma^2_{\phi,i}(\mathbf{x}) + \mu^2_{\phi,i}(\mathbf{x}) - 1 - \log\sigma^2_{\phi,i}(\mathbf{x})\right) + C. \tag{49}$$

3. Forward Pass in DDPMs:

$$p(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}) \tag{50}$$

$$p(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I}) \tag{51}$$

4. ϵ-Prediction loss in DDPMs:

$$\mathcal{L}(\theta) = \mathbb{E}_{t,\mathbf{x}_0,\epsilon}\left[||\epsilon_\theta(\mathbf{x}_t, t) - \epsilon||^2\right]. \tag{52}$$

5. $\mathbf{x}_0$-Prediction loss in DDPMs:

$$\mathcal{L}(\theta) = \mathbb{E}_{t,\mathbf{x}_0,\epsilon}\left[\omega_t||\mathbf{x}_\theta(\mathbf{x}_t, t) - \mathbf{x}_0||^2\right]. \tag{53}$$

4

ENERGY-BASED MODELS

Energy-Based Models (EBMs) [30] are a general class of models that capture dependencies by associating a scalar energy value to each configuration of variables, acting as a measure of compatibility.

- **Training**: The general principle is to find an energy function that associates low energies to correct (observed) configurations of variables, and higher energies to incorrect (unobserved) configurations.
- **Inference**: Making a decision consists of setting the values of observed variables and finding values for the missing variables that minimize the energy, i.e., finding the values most compatible with the observation. In the simplest scenario with two variables X and Y, where the first is observed, the model will produce an answer Y^* that is the most compatible with X. This procedure is formally defined as:

$$Y^* = \text{argmin}_{Y \in \mathcal{Y}} E(X, Y). \tag{54}$$

  When the size of the set $\mathcal{Y}$ is small, we can simply compute the energy for all possible values and pick the smallest. However, we typically deal with complex, high-dimensional spaces where exhaustive search is impractical. The solution is to apply an inference procedure to obtain an approximate result, which may or may not be the global minimum of the energy function. For example, for a continuous variable Y and a smooth, well-behaved energy function, gradient-based optimization algorithms can be utilized.

The scenario described above covers prediction, classification, and decision-making, answering the question: "Which value of Y is most compatible with this X?". One advantage of these models is that we can use energy-based models to answer different types of questions. (I) Ranking, "Is Y_1 or Y_2 more compatible with this X?"; (II) Detection, "Is this value of Y compatible with X?"; and (III) Conditional density estimation, "What is the conditional probability distribution over $\mathcal{Y}$ given X?".

In deterministic settings, we only require the system to assign the lowest energy to the correct value, allowing us to ignore the absolute scale of the remaining large energies. However, if these outputs need to be rigorously compared, combined, or utilized to quantify uncertainty, uncalibrated energies are insufficient. We can calibrate ener-

gies by transforming them into a normalized probability distribution using the Gibbs distribution:

$$p_\theta(\mathbf{x}) = \frac{e^{-E_\theta(\mathbf{x})}}{Z_\theta}, \qquad Z_\theta = \int_{\mathbf{x}\in\mathcal{X}} e^{-E_\theta(\mathbf{x})} d\mathbf{x} \tag{55}$$

where the denominator Z_θ is called the *partition function* ensuring normalization:

$$\int_{\mathbf{x}\in\mathcal{X}} p_\theta(\mathbf{x}) d\mathbf{x} = 1. \tag{56}$$

Under this framework, lower energy values correspond to higher probabilities and the partition function Z_θ is a normalisation constant that ensures the probabilities sum to one.

We can train probabilistic EBMs using Maximum Likelihood Estimation (MLE), defined by minimizing the negative log-likelihood of the empirical data:

$$\mathcal{L}(\theta) = -\mathbb{E}_{\mathbf{x}\sim p_{data}}[\log p_\theta(\mathbf{x})] = -\mathbb{E}_{\mathbf{x}\sim p_{data}}\left[\log \frac{e^{-E_\theta(\mathbf{x})}}{Z_\theta}\right] \tag{57}$$

$$= \mathbb{E}_{\mathbf{x}\sim p_{data}}[E_\theta(\mathbf{x})] + \log Z_\theta \tag{58}$$

where Z_θ is the partition function defined in Equation –. The first term encourages the model to assign low energy to data points, while the second enforces normalization. The problem is that the partition function is intractable to compute for high-dimensional data, which makes the training of EBMs challenging. To overcome this, the EBM literature relies on two broad families of approximations: contrastive methods (such as Markov Chain Monte Carlo or Contrastive Divergence [11]), which approximate the model expectation, and non-contrastive/regularized methods (such as Score Matching [15] or Denoising Autoencoders), which bypass the partition function entirely by restructuring the objective.

In statistics, the score function of a probability distribution $p(\mathbf{x})$ is defined as the gradient of the log-probability density with respect to the input space $\mathbf{x}$:

$$\mathbf{s}(\mathbf{x}) = \nabla_\mathbf{x} \log p(\mathbf{x}) \tag{59}$$

When we apply this definition to the Gibbs distribution of our Energy-Based Model, a remarkable mathematical cancellation occurs:

$$\mathbf{s}\theta(\mathbf{x}) = \nabla\mathbf{x} \log p_\theta(\mathbf{x}) \tag{60}$$

$$= \nabla_\mathbf{x}(-E_\theta(\mathbf{x}) - \log Z_\theta) \tag{61}$$

$$= -\nabla_\mathbf{x} E_\theta(\mathbf{x}) \tag{62}$$

Because the partition function Z_θ is a constant with respect to the input $\mathbf{x}$, its gradient vanishes ($\nabla_{\mathbf{x}} \log Z_\theta = 0$). Consequently, the score of an EBM is simply the negative gradient of its energy function. This allows us to evaluate the vector field pointing toward regions of higher probability without ever calculating the normalization constant.

4.1 SCORE MATCHING

4.2 DENOISING SCORE MATCHING

Summary of Key Equations: EBMs and Score Matching

This box summarizes the main mathematical equations of EBMs and Score Matching:

1. Training Objective of VAEs:

$$\mathcal{L}(\theta, \phi; \mathbf{x}) = \mathbb{E}_{z \sim q_\phi(\mathbf{z}|\mathbf{x})} [\log p_\theta(\mathbf{x}|\mathbf{z})] - D_{KL}(q_\phi(\mathbf{z}|\mathbf{x}) || p_\theta(\mathbf{z})) \quad (63)$$

2. Reparameterization Trick:

$$\mathbf{x}_{\text{synthetic}} = G(\mathbf{z}_{\text{anatomy}} \oplus \mathbf{z}_{\text{pathology}} \mid \mathbf{y}) \quad (64)$$

Note: Ensure latent dimensions z_i and z_j maintain near-zero mutual information to preserve explainability.

5 NORMALIZING FLOWS

Normalizing Flows.

6

DISENTANGLEMENT

Disentangled Representation Learning (DRL) is a paradigm in which models are designed to extract and separate the hidden factors of variation, $\mathbf{z} \in \mathbb{R}^c$, from an input observation, $\mathbf{x} \in \mathbb{R}^d$. These generative factors are commonly referred to as latent variables. As defined in [1], a "disentangled representation should separate the distinct, independent and informative generative factors of variation in the data. Single latent variables are sensitive to changes in single underlying generative factors, while being relatively invariant to changes in other factors."
This definition highlights two core principles: the statistical independence of the latent variables, and the guarantee that modifying one factor alters only its corresponding visual concept, leaving all others unchanged. Organizing the latent space into these explainable semantic representations enhances a generative model in three key ways:

- **Explanability**: the meaningful and independent representations are aligned semantically with latent generative factors.
- **Generalizability**: the representations are well separated from the original entangled input, which improves the generalization ability for the generation of new samples.
- **Controllability**: DRL allows for controllable generation by manipulating the learned disentangled representations of the latent space.

Current literature features various disentanglement methods. As discussed in the following sections, these approaches are typically categorized based on their underlying generative framework [50].

While the theoretical benefits of DRL are significant, achieving completely unsupervised disentanglement presents a fundamental challenge known as the identifiability problem. As demonstrated in [33], learning disentangled representations in a purely unsupervised manner is theoretically impossible without introducing explicit inductive biases on both the models and the data.
The core of this problem lies in statistical identifiability. Unsupervised generative models aim to learn the true prior distribution of the latent factors, $p(\mathbf{z})$, by observing only the marginal distribution of the data, $p(\mathbf{x})$. However, [33] formally proved that there exists an infinite number of bijective transformations, $f(\mathbf{z})$, that are highly entangled but produce the exact same marginal distribution $p(\mathbf{x})$. Because the

model only has access to the observational data, it has no mathematical way to distinguish between the true, disentangled latent space and an entangled, transformed equivalent. Consequently, the true generative factors are unidentifiable.
To overcome this fundamental limitation, current methodologies must shift away from purely unsupervised approaches and incorporate specific constraints. This is typically achieved through three primary strategies:

- **Inductive Biases**: Imposing strong architectural constraints or specific regularization terms (e.g., heavily penalizing total correlation) to favor disentangled solutions, assuming the data possesses a specific underlying structure.
- **Weak Supervision**: Introducing limited supervisory signals, such as providing a small number of labeled samples or using paired observations that share at least one underlying generative factor (e.g., two images of the same object under different lighting).
- **Sequential and Temporal Dynamics**: Leveraging the natural continuity found in sequential data, such as video frames, where most latent factors remain constant between consecutive observations while only a few change dynamically.

6.1 VAE-BASED MODELS

In Section **??** we defined the Variational Autoencoder (VAE) model, where the loss function is defined in Equation **??**. This type of model has the potential to learn disentangled representations in simple datasets, because of the choose of the prior distribution $p(\mathbf{z})$ as a normal distribution, which imposes independence constraints. In more complex datasets, the disentanglement capability presented in this model is poor.

One of the first methods was **β-VAE** [10], which introduces a β penalty hyperparameter before the KL term in the ELBO loss. The new training objective is defined as

$$\mathcal{L}_{\text{ELBO}}(\theta, \phi; \mathbf{x}) = \mathbb{E}_{\mathbf{z}\sim q_\phi(\mathbf{z}|\mathbf{x})}\left[\log p_\theta(\mathbf{x}|\mathbf{z})\right] - \beta D_{KL}(q_\phi(\mathbf{z}|\mathbf{x})||p(\mathbf{z})), \quad (65)$$

where a value of $\beta = 1$ corresponds to the original VAE implementation and larger values of $\beta > 1$ encourage learning more disentangled representations, adding more pressure to the latent bottleneck to be more factorised, while reducing the performance of reconstruction. This shows an important trade-off between reconstruction accuracy and the disentanglement quality of the latent representation.

Several studies have analized this trade-off and the effect of the β

hyperparameter in the disentanglement performance. First, we can define the marginal posterior distribution as

$$q_\phi(\mathbf{z}) = \mathbb{E}_{p_{data}(\mathbf{x})}\left[q_\phi(\mathbf{z}|\mathbf{x})\right] = \int q_\phi(\mathbf{z}|\mathbf{x})p_{data}(\mathbf{x})d\mathbf{x}, \tag{66}$$

where a disentangled representation is achieved when $q_\phi(\mathbf{z})$ is close to the factorial prior $p(\mathbf{z})$, which means we want a factorial distribution $q_\phi(\mathbf{z}) = \prod_j q_\phi(z_j)$. To gain a deep insight about the trade-off, we can break down the KL term of the loss function as proposed in several papers [4, 14]

$$\begin{aligned}\mathbb{E}_{p_{data}(\mathbf{x})}\left[KL(q_\phi(\mathbf{z}|\mathbf{x})||p(\mathbf{z}))\right] = & \underbrace{I(\mathbf{x};\mathbf{z})}_{\text{(i) Mutual Information}} \\ & + \underbrace{D_{KL}(q_\phi(\mathbf{z})||\bar{q}_\phi(\mathbf{z}))}_{\text{(ii) Total Correlation}}, \\ & + \underbrace{D_{KL}(q_\phi(\mathbf{z})||p(\mathbf{z}))}_{\text{(iii) Prior KL}},\end{aligned} \tag{67}$$

where $I(\mathbf{x};\mathbf{z})$ is the mutual information between $\mathbf{x}$ and $\mathbf{z}$, Total Correlation (TC) is a measure of dependence between the variables and the prior KL prevents the latents to deviate too far from the prior. Here we can clearly see that making β larger pushes $q_\phi(\mathbf{z})$ towards the factorial prior $p(\mathbf{x})$ and pushes the model to find statistically independent factors, improving the disentanglement, but reduces the amount of information about $\mathbf{x}$ stored in $\mathbf{z}$, producing a poor reconstruction.

This trade-off between reconstruction and disentanglement is the main motivation for the development of new methods based on the previous decomposition of the KL term. For example, the **DIP-VAE-I** and **DIP-VAE-II** [27] methods proposed a regularization term on the third element of the KL decomposition. The objective function can be defined as

$$\begin{aligned}\mathcal{L}_{ELBO}(\theta,\phi;\mathbf{x}) = & \mathbb{E}_{\mathbf{z}\sim q_\phi(\mathbf{z}|\mathbf{x})}\left[\log p_\theta(\mathbf{x}|\mathbf{z})\right] - D_{KL}(q_\phi(\mathbf{z}|\mathbf{x})||p(\mathbf{z})) \\ & - \lambda D_{KL}(q_\phi(\mathbf{z})||p(\mathbf{z})).\end{aligned} \tag{68}$$

This objective function imposes independence on the variational marginal distribution $q_\phi(\mathbf{z})$. To minimize the distance, they force the covariance of $q_\phi(\mathbf{z})$ to be close to the identity matrix. The covariance is defined as

$$Cov_{q_\phi(\mathbf{z})}[\mathbf{z}] = \mathbb{E}_{p_{data}(\mathbf{x})}\left[\Sigma_\phi(\mathbf{x})\right] + Cov_{p_{data}(\mathbf{x})}\left[\mu_\phi(\mathbf{x})\right]. \tag{69}$$

The two variations of DIP-VAE differ in the way they penalise the covariance:

$$\begin{aligned} &D_{KL}(q_\phi(\mathbf{z})\|p(\mathbf{z})) = \\ &\underbrace{= \lambda_{od} \sum_{i \neq j} [\mathrm{Cov}_{p_{data}}[\mu_\phi(\mathbf{x})]]_{ij}^2 + \lambda_d \sum_i \left(\mathrm{Cov}_{p_{data}}[\mu_\phi(\mathbf{x})]_{ij} - 1\right)^2}_{\text{DIP-VAE-I}}, \\ &\text{or} \\ &\underbrace{= \lambda_{od} \sum_{i \neq j} [\mathrm{Cov}_{q_\phi(\mathbf{z})}[\mathbf{z}]]_{ij}^2 + \lambda_d \sum_i \left([\mathrm{Cov}_{q_\phi(\mathbf{z})}[\mathbf{z}]]_{ij} - 1\right)^2}_{\text{DIP-VAE-II}}, \end{aligned} \tag{70}$$

where λ_{od} and λ_d are the hyperparameters that control the strength of the regularization.

The next method, **FactorVAE** [21], proposed to only penalises the second component of the KL decomposition, the Total Correlation (TC), which encourage independence in the latent space. The loss function in this method is defined as:

$$\begin{aligned} \mathcal{L}_{ELBO}(\theta, \phi; \mathbf{x}) = \mathbb{E}_{\mathbf{z} \sim q_\phi(\mathbf{z}|\mathbf{x})} [\log p_\theta(\mathbf{x}|\mathbf{z})] - D_{KL}(q_\phi(\mathbf{z}|\mathbf{x})\|p(\mathbf{z})) \\ - \gamma D_{KL}(q_\phi(\mathbf{z})\|\bar{q}_\phi(\mathbf{z})), \end{aligned} \tag{71}$$

where $\bar{q}_\phi(\mathbf{z}) = \prod_j q_\phi(z_j)$. TC measures the dependencies between the variables in the latent representation. Penalising this specific term encourages the latent dimensions to be statistically independent, which is the core mathematical interpretation of disentanglement.
Because the Total Correlation is computationally intractable to evaluate directly over the entire dataset, FactorVAE employs an adversarial training approach. A discriminator network is trained to distinguish between samples drawn from the aggregated posterior $q_\phi(\mathbf{z})$ and samples drawn from the product of its marginals $\prod_j q_\phi(z_j)$, effectively estimating and minimizing the density ratio. This approach achieves better disentanglement than β-VAE while preserving a higher reconstruction quality.

The paper that proposed the decomposition of the KL term [4] also proposed a method called β-**TCVAE**, which penalize all the terms of the KL decomposition independently, with different hyperparameters. The loss function is defined as

$$\begin{aligned} &\mathcal{L}_{ELBO}(\theta, \phi; \mathbf{x}) = \mathbb{E}_{\mathbf{z} \sim q_\phi(\mathbf{z}|\mathbf{x})} [\log p_\theta(\mathbf{x}|\mathbf{z})] \\ &- \alpha I(\mathbf{x}; \mathbf{z}) - \beta D_{KL}(q_\phi(\mathbf{z})\|\bar{q}_\phi(\mathbf{z})) - \gamma D_{KL}(q_\phi(\mathbf{z})\|p(\mathbf{z})), \end{aligned} \tag{72}$$

All the previous methods are unsupervised with the common idea of adding extra regularization terms to the original VAE loss function to encourage disentanglement.

6.2 DIFFUSION / FLOW-BASED MODELS

Diffusion and Flow models are a more recent and advanced class of generative models that have drawn a lot of attention in recent years because of their remarkable performance. Disentangled representation learning can also be applied to these types of models with the aim to overcome the important trade-off between reconstruction and disentanglement of the VAE-based methds.

One of the first approaches was DisDiff [53], which learns a disentangled representation and a disentangled gradient field for each generative factor. Assume the data samples are generated by N underlying ground truth factors $\mathbf{f}^c$, $c \in \{1, \dots, N\}$. Each factor c follows the distribution $p(\mathbf{f}^c)$. We can modify the original score function (Equation –) with the conditioning of the factors using the Bayes' Rule as follows:

$$\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t|\mathbf{f}^c) = \nabla_{\mathbf{x}_t} \log p(\mathbf{f}^c|\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t), \tag{73}$$

where we already have the score $\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t)$ from a pretrained unconditional model and we only have to learn $\nabla_{\mathbf{x}_t} \log p(\mathbf{f}^c|\mathbf{x}_t)$. Since we are in an unsupervised setting, we do not know $\mathbf{f}^c$. We have to learn a set of representations $\{\mathbf{z}^c|c = 1, \dots, N\}$ that must encompass all information of the original samples $\mathbf{x}_0$ and represent the true factors of variations $\mathbf{f}^c$. To obtain these representations we use a set of encoders with learnable parameters ϕ, defined as $E_\phi(\mathbf{x}_0) = \{E^1_\phi(\mathbf{x}_0), \dots E^N_\phi(\mathbf{x}_0)\} = \{\mathbf{z}^1, \dots, \mathbf{z}^N\}$. Then, assume we have a factor subset $S \subseteq C$ with $\mathbf{z}^S = \{\mathbf{z}^c|c \in S\}$, the gradient field can be defined as

$$\nabla_{\mathbf{x}_t} \log p(\mathbf{z}^S|\mathbf{x}_t) = \sum_{c \in S} \nabla_{\mathbf{x}_t} \log p(\mathbf{z}^c|\mathbf{x}_t). \tag{74}$$

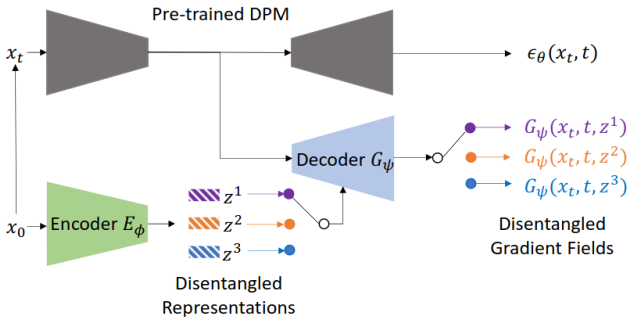
A visual representation of this method can be seen in Figure **??**(a). Finally, to estimate the gradient fields, they used a network $G_\psi(\mathbf{x}_t, \mathbf{z}^c, t)$ and the reconstruction loss is defined as

$$\mathcal{L}_r = \mathbb{E}_{\mathbf{x}_0,t,\epsilon}\|\epsilon - \epsilon_\theta(\mathbf{x}_t, t) + \frac{\sqrt{\alpha_t}\sqrt{1-\bar{\alpha}_t}}{\beta_t}\sigma_t \sum_{c \in C} G_\psi(\mathbf{x}_t, \mathbf{z}^c, t)\|. \tag{75}$$

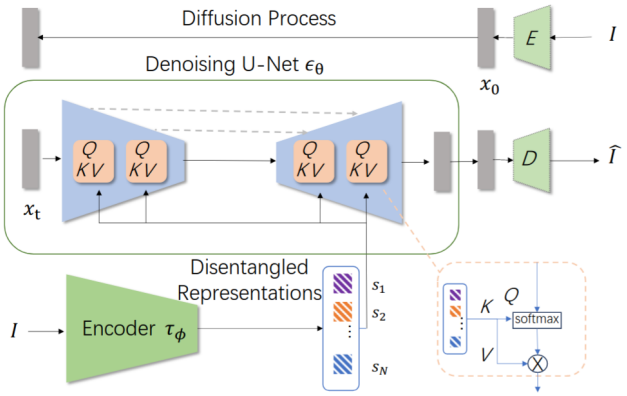
This loss formulation alone does not guarantee disentanglement among the representations $\mathbf{z}^c$. The authors proposed a disentanglement loss that minimizes the upper bound for the mutual information between $\mathbf{z}^c$ and $\mathbf{z}^j$, where $c \neq j$, defined as

$$\mathcal{L}_{dis} = \min_{i,j,x_0,\hat{\mathbf{x}}^j_0} \|\hat{\mathbf{z}}^{c|j} - \hat{\mathbf{z}}^c\| - \|\hat{\mathbf{z}}^{c|j} - \mathbf{z}^c\| + \|\hat{\mathbf{z}}^{j|j} - \mathbf{z}^j\|, \tag{76}$$

where $\hat{\mathbf{x}}^j_0$ is conditioned on $\mathbf{z}^j$, $\hat{\mathbf{z}}^c = E^c_\phi(\hat{\mathbf{x}}_0)$ and $\hat{\mathbf{z}}^{c|j} = E^c_\phi(\hat{\mathbf{x}}^j_0)$ where $\hat{\mathbf{x}}_0$ is a sampled data from the pretrained model.



(a) DisDiff Framework



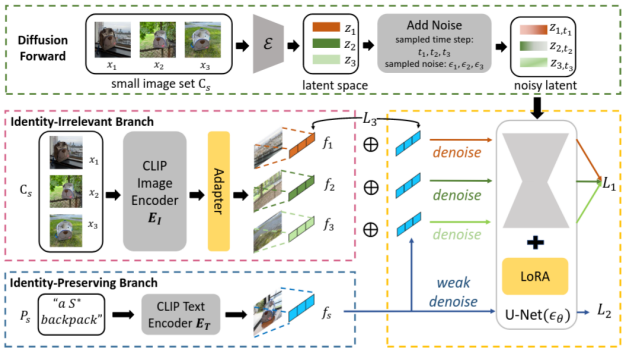
(b) EncDiff Framework

Figure 5: Illustrations of the frameworks for the (a) DisDiff model [53] and the (b) EncDiff model. [52]

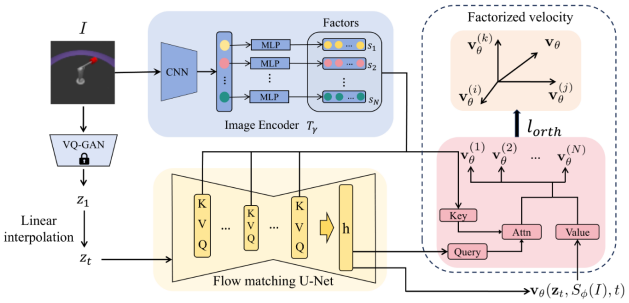
EncDiff [52] is an improvement of the previous method where an image encoder τ_ϕ aims to provide a set of concept tokens $\mathcal{S} = \{s_1, \ldots, s_N\}$. Each dimension of the encoded feature vectors is treated as a disentangled factor and maps each to a vector using a non-shared MLP layer. Then, based on Latent Diffusion Models [43], the idea is to use the cross-attention mechanism to map the tokens to the intermediate representations of the U-Net model. The general framework of this method is illustrated in Figure **??**(b).

DisenBooth [3]

After the development of several diffusion-based disentanglement methods, flow matching with good performance and deterministic formulation has gained attention in this field. In a recent paper, a Flow Matching-Based disentanglement framework [6] was presented that encourages factor-specific, non-overlapping latent transformations and explicit semantic alignment. The pipeline of this method can be seen in Figure **??**(b). Given an image $\mathbf{I}$, they encode it into a latent $\mathbf{Z}$ using a pre-trained VQ-GAN encoder, where $\mathbf{z}_1$ represents the latent of an image of the dataset. Then, a factor encoder $(T_\gamma(\mathbf{x}))$ is trained to



(a) DisenBooth Framework



(b) FM Disentanglement Framework

Figure 6: Illustrations of the frameworks for the (a) DisenBooth model [3] and the (b) FM Disentanglement model. [6]

extract a set of N factors. These factors are injected into the flow network via cross-attention, where intermediate features of the network attend to the set of factors. The model outputs a factor-conditioned velocity field $\mathbf{v}_\theta(\mathbf{z}, T_\gamma(\mathbf{x}), t)$. Since the flow matching loss only takes into account the velocity field, it does not encourage disentanglement by default. The next step is to decompose the velocity into N factor-aligned components.
The idea is to decompose the factor-conditioned velocity field as follows:

$$\mathbf{v}_\theta(\mathbf{z}_t, T_\gamma(\mathbf{x}), t) = \sum_{i=1}^{N} \mathbf{v}_\theta^{(i)}(\mathbf{z}_t, T_\gamma(\mathbf{x}), t) \tag{77}$$

where $\mathbf{v}_\theta^{(i)}$ is the component attributed to the i-th factor of variation. They add an orthogonality loss to penalise pairwise alignment between factor-specific components and encourage different factors to capture distinct transport directions. For the definition of the different velocity fields, they used output attention. Let h denote the last hidden feature map of the network, we define the queries and keys as:

$$\mathbf{Q} = \mathbf{W}_q h, \quad \mathbf{K}_i = \mathbf{W}_k s_i \tag{78}$$

where $\mathbf{W}_q$ and $\mathbf{W}_k$ are learned projections, s_i a factor, and d the query/key dimensionality. The attention is defined as

$$\mathrm{Attn}_i = \frac{\exp\left(\frac{\mathbf{Q}\mathbf{K}_i^T}{\sqrt{d}}\right)}{\sum_{j=1}^{N} \exp\left(\frac{\mathbf{Q}\mathbf{K}_i^T}{\sqrt{d}}\right)}, \tag{79}$$

which satisfies that $\sum_{i=1}^{N} \mathrm{Attn}_i = 1$ at each spatial location. Then, the factor-specific velocity field is defined as

$$\mathbf{v}_\theta^{(i)}(\mathbf{z}_t, T_\gamma(\mathbf{x}), t) = \mathrm{Attn}_i \odot \mathbf{v}_\theta(\mathbf{z}_t, T_\gamma(\mathbf{x}), t), \tag{80}$$

where $\odot$ denotes the element-wise multiplication. The final training objective is defined as

$$\mathcal{L}(\theta) = \mathcal{L}_{FM}(\theta) + \lambda_{orth}\mathcal{L}_{orth} \tag{81}$$

where λ_{orth} controls the strength of the regularization.

6.3 METRICS

One of the initial methods for the evaluation of disentanglement representation learning was inspecting the change in reconstruction when traversing one variable in the latent space. This qualitative evaluation is useful, but it is important to have reliable quantitative metrics to measure disentanglement. As shown in the previous sections, significant advances have been made in this field, but the literature still lacks a reliable and unified metric for evaluation. As presented in [18], a good metric has to evaluate the separation of interpretable factors, the capture of the diversity of the data, the invariance across changes in unrelated dimensions and the summarisation of the important information efficiently in a disentangled representation.

One of the initial metrics presented in the literature are **Z-diff** (also called BetaVAE) [10] and **Z-min Variance** (also called FactorVAE) [21], where both of them are supervised, i.e. we need the ground truth factors of variations. The idea is to choose a factor k, generate data with this factor fixed and the others varying randomly, obtain the representations through the encoder ($q_\phi(\mathbf{z}|\mathbf{x})$) and take the absolute value of the pairwise differences. In Z-diff, the metric trains a classifier to identify which factor is fixed and reports the accuracy value as the disentanglement metric. On the other hand, metric Z-min Variance shows that the previous one has some problems related to the training of a model and proposes to avoid training a classifier and use a majority-vote classifier with no optimisation hyperparameters.

Then, we have an information-theoretic metric called **Mutual Information Gap** (MIG) [4]. The idea is to evaluate the empirical mutual

information between the latent variable $\mathbf{z}_j$ and ground truth factor $\mathbf{v}_k$, defined as $I(\mathbf{z}_j; \mathbf{v}_k)$. A higher mutual information implies that $\mathbf{z}_j$ contains a lot of information about $\mathbf{v}_k$. For each real factor $\mathbf{v}_k$, the metric computes the gap between the top two latent variables with the highest mutual information, and the average over all factors is the reported metric, defined as

$$\text{MIG} = \frac{1}{K} \sum_{k=1}^{K} \frac{1}{H(\mathbf{v}_k)} \left(I(\mathbf{z}_{j^{(k)}}; \mathbf{v}_k) - \max_{j \neq j^{(k)}} I(\mathbf{z}_j; \mathbf{v}_k) \right), \tag{82}$$

where $j^{(k)} = \arg\max_j I(\mathbf{z}_j; \mathbf{v}_k)$ and K is the number of known factors. The metric is bounded by 0 and 1, with higher values indicating better disentanglement performance. This means that each generative factor is primarily captured by only one latent dimension.
Later, some variations of the MIG metrics were proposed in the literature. MIG-sup [31] propose also to compute the mutual information gap but for the different latent dimensions, which is defined as

$$\text{MIG-sup} = \frac{1}{J} \sum_{j=1}^{J} \left(I(\mathbf{z}_j; \mathbf{v}_{k^{(j)}}) - \max_{k \neq k^{(j)}} I(\mathbf{z}_j; \mathbf{v}_k) \right), \tag{83}$$

where J is the number of latent dimensions.

A different disentanglement metric presented in the literature is **Separated Attribute Predictability** (SAP), which constructs a score matrix $\mathbf{S} \in \mathbb{R}^{d \times k}$ and the (i, j)-th element represents the linear regression or classification score of predicting j-th generative factor using only the i-th latent variable. For each column of the matrix, which corresponds to a generative factor, we take the difference of the top two entries (corresponding to the top two most predictive latent dimensions), and then take the mean of these differences as the final SAP value. Higher scores mean better disentanglement performance because it indicates that each generative factor principally corresponds to only one latent dimension, similar to MIG.

The authors of [8] proposed a framework to evaluate disentanglement from three aspects, disentanglement (D), completeness (C) and informativeness (I), to build the metric called **DCI**. For the evaluation of the aspects, we have to train K regressors, where each one f_j predicts $\mathbf{z}_j$ given $\mathbf{c}$ and can provide an importance score matrix $\mathbf{R}_{i,j}$ where the (i, j)-th elements denotes the relative importance of $\mathbf{c}_i$ in predicting $\mathbf{z}_j$. Once we have the regressors:

- **Disentanglement**. It is defined as the degree to which a representation factorises the underlying factors of variation. The disentanglement score D_i of the code variable $\mathbf{c}_i$ is quantified by $D_i = (1 - H_K(P_{i\cdot}))$, where $H_K(P_{i\cdot}) = -\sum_{k=0}^{K-1} P_{ik} \log_K P_{ik}$

denotes the entropy and $P_{ij} = R_{ij} / \sum_{k=0}^{K-1} R_{ik}$ denotes the probability of $\mathbf{c}_i$ being important for predicting $\mathbf{z}_j$. If it is important, the score will be 1. If i is equally important for predicting all generative factors, the metric will be 0.

- **Completeness**. It is the degree to which each underlying factor is captured by a single code variable. The completeness score C_j in capturing the generative factor $\mathbf{z}_j$ is defined as $C_j = (1 - H_D(\tilde{P}_{\cdot j}))$, where $H_D(\tilde{P}_{\cdot j}) = -\sum_{d=0}^{D-1} \tilde{P}_{dj} \log_D \tilde{P}_{dj}$. If a single latent variable contributes to the prediction of $\mathbf{z}_j$, the score will be 1.

- **Informativeness**. It is the amount of information that a representation captures about the underlying factors of variations. The informativeness of the latent $\mathbf{c}$ about the generative factor $\mathbf{z}_j$ is quantified by the prediction error $E(\mathbf{z}_j, \hat{\mathbf{z}}_j)$, averaged over the dataset. It is important to mention that this metric is dependent on the capacity of the regressor models f.

One recent metric was designed after the analysis of the previous metrics, which reflects the desired properties and is experimentally more robust. The metric is called **EDI** [18] and the idea is to compute the disentanglement (D), the compactness (C) and the exclusivity (E). First, we have to compute the relationship matrix, where we have to calculate $I(\mathbf{c}_i, \mathbf{z}_j)$, the mutual information between latent $\mathbf{c}_i$ and ground truth factor $\mathbf{z}_j$, then $I(\mathbf{c}_1, \dots, \mathbf{c}_d, \mathbf{z}_j)$, the mutual information between all latent codes and the factor $\mathbf{z}_j$, and $H(\mathbf{z}_j)$, the entropy of each factor. The impact intensity of factor $\mathbf{z}_j$ on the latent code $\mathbf{c}_i$ among all codes is defined as

$$R(\mathbf{c}_i, \mathbf{z}_j) = \frac{I(\mathbf{c}_i, \mathbf{z}_j)}{I(\mathbf{c}_i, \dots, \mathbf{c}_d, \mathbf{z}_j)}. \tag{84}$$

Then, another important concept is exclusivity, which can be defined as

$$\text{Exclusivity}(\mathbf{a}_1, \dots, \mathbf{a}_n) = \mathbf{a}_{i^*} - \sqrt{\frac{1}{n-1} \sum_{i=1, i \neq i^*}^{n} \mathbf{a}_i^2}, \tag{85}$$

where $i^* = \arg\max_i \mathbf{a}_i$ and the aim is for the maximum value to be as high as possible, with the remainder as minimal as possible. Now, we can define the metrics used in the EDI evaluation framework:

- **Disentanglement**. The disentanglement of a latent code $\mathbf{c}_i$ can be defined as

$$D(\mathbf{c}_i) = \text{Exclusivity}(R(\mathbf{c}_i, \mathbf{z}_1), R(\mathbf{c}_i, \mathbf{z}_2), \dots, R(\mathbf{c}_i, \mathbf{z}_k)). \tag{86}$$

- **Compactness**. The compactness of a generative factor $\mathbf{z}_j$ is defined as

$$C(\mathbf{z}_j) = \text{Exclusivity}(R(\mathbf{c}_1, \mathbf{z}_j), R(\mathbf{c}_2, \mathbf{z}_j), \dots, R(\mathbf{c}_d, \mathbf{z}_j)). \tag{87}$$

- **Explicitness**. For a generative factor $\mathbf{z}_j$, the explicitness is calculated as the ratio of the combined information of all the codes relative to $\mathbf{z}_j$ to the entropy of $\mathbf{z}_j$ itself, defined as

$$I(\mathbf{z}_j) = \frac{I(\mathbf{c}_i, \mathbf{c}_2, \ldots, \mathbf{c}_d, \mathbf{z}_j)}{H(\mathbf{z}_j)}. \tag{88}$$

7

COMPOSITIONALITY

Compositionality.

7.1 OBJECT-CENTRIC LEARNING

Slot Attention (SA) [34] is a differentiable interface between image features and a set of variables called slots. This method maps a set of N image patches with dimension D_{input}, called image features $\mathbf{F} \in \mathbb{R}^{N \times D_{input}}$, to a set of K output vectors of size D_{slot} called slots, $\mathbf{S} \in \mathbb{R}^{K \times D_{slot}}$. Each vector of the slots can describe an object or element in the input image. The process of binding a part of the image to the slots is performed via iterative refinement over T steps, where in each iteration the slots compete via a softmax attention mechanism to explain parts of the input. Slot representations are initialised randomly with a Gaussian distribution.
If we define some learnable linear transformations k, q and v to map input features and slots to a common dimension D, we can define:

$$\mathbf{A}_{i,j} = \frac{e^{\mathbf{M}_{i,j}}}{\sum_l e^{\mathbf{M}_{i,l}}}, \text{ where } \mathbf{M} = \frac{1}{\sqrt{D}} k(\mathbf{F}) \cdot q(\mathbf{S})^T \in \mathbb{R}^{N \times K} \tag{89}$$

The normalisation in the attention ensures that attention coefficients sum to one for each individual input feature vector. The update for the slots can be defined as:

$$\mathbf{U} = \mathbf{W}^T \cdot v(\mathbf{F}) \in \mathbb{R}^{K \times D} \text{ where } \mathbf{W}_{i,j} = \frac{\mathbf{A}_{i,j}}{\sum_{l=1}^{N} \mathbf{A}_{l,j}}. \tag{90}$$

Slot updates are performed via a Gated Recurrent Unit (GRU) and a small MLP layer with ReLU and residual connections, applied independently to each slot using shared parameters.
Once the slot representations are learned, they can be used for several downstream tasks. Some examples in the original paper [34] are

- **Object discovery**: SA can be used as a part of an autoencoder model for unsupervised object discovery. The encoder encodes the image in a set of slots, and the decoder reconstructs the original image. In this case, the slots act as a representation bottleneck. They proposed the use of a spatial broadcast decoder, which removes the upsampling deconvolutions. This new decoder first broadcasts the latent vector across the space (height and width of the image), appends fixed coordinate channels and applies a fully convolutional network to produce a final tensor with 4 channels, 3 for RGB and the alpha channel.

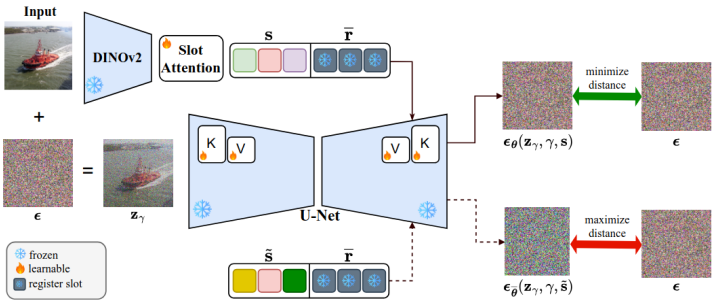


Figure 7: Overview of the pipeline of the CODA method.

- **Set prediction**: In this scenario, we have an input image and a set of prediction targets. The main challenge is that the output order in SA is random, so there are K! possible equivalent representations for a set of K slots. In this case, they proposed to apply for each slot an MLP with shared parameters between slots. To overcome the problem of different prediction and label orders, they used a matching algorithm in the loss computation.

The next step is the **Contrastive Object-centric Diffusion Alignment** (CODA) [38], a diffusion-based OCL where the slots ($\mathbf{s}$) are extracted using Slot Attention over DINOv2 features and decoded with a pretrained Stable Diffusion v1.5 model. An illustration of the pipeline is presented in Figure **??**. This method introduces three key components to improve the disentanglement and the alignment between the slot and the visual components of the image.

- **Register Slots**. The idea is to define a set of input-independent register slots to capture the residual information, such as the background, which is usually entangled with all the slots. They proposed this method based on some experiments where reconstruction from the full set of slots resembles the input image, but using only a small set of slots, the result is a distorted image, which is not consistent with an ideal OCL model. These register slots ($\bar{\mathbf{r}}$) are defined as frozen embeddings by encoding with a pretrained text encoder a sequence of `PAD` tokens.

- **Finetune of Cross-Attention**. Since we are using a pre-trained SD model trained on large-scale image-text pairs, using the slot embeddings directly will introduce a bias, as it expects text embeddings. They proposed to only finetune the key, value, and output projections in cross-attention layers, which allows the model to better align slots with visual content and preserve the expressiveness of the pretrained diffusion backbone. Also, this approach is computationally and memory efficient.

- **Contrastive Alignment Loss**. Using only the diffusion loss, the model does not have explicit supervision to ensure that slots

capture concepts present in the image. They proposed a contrastive alignment objective that aligns slots with image content while discouraging overlap between different slots. The idea is to assign high likelihood to the correct slot representations and low likelihood to negative slots ($\tilde{\mathbf{s}}$), which are created as a random shuffle between slots of two different images.

The overall training objective of CODA is defined as

$$\mathcal{L}(\phi, \theta) = \mathcal{L}_{dm}(\phi, \theta) + \lambda_{cl}\mathcal{L}_{cl}(\phi), \tag{91}$$

where ϕ are the parameters of Slot Attention, θ the parameters of the diffusion models and λ_{cl} controls the trade-off between the denoising and contrastive terms. The losses are defined as

$$\mathcal{L}_{dm}(\phi, \theta) = \mathbb{E}_{\epsilon, t, (\mathbf{x}, \mathbf{s})}\left[||\epsilon - \epsilon_\theta(\mathbf{x}_t, t, (\mathbf{s}, \bar{\mathbf{r}}))||_2^2\right] \tag{92}$$

$$\mathcal{L}_{cl}(\phi) = -\mathbb{E}_{\epsilon, t, (\mathbf{x}, \tilde{\mathbf{s}})}\left[||\epsilon - \epsilon_\theta(\mathbf{x}_t, t, (\tilde{\mathbf{s}}, \bar{\mathbf{r}}))||_2^2\right] \tag{93}$$

8

CONCEPT BOTTLENECK MODELS

Concept Bottleneck Models (CBMs) were first introduced in [25] with the idea to easily interact with state-of-the-art models using high-level concepts that are understandable for the researchers. The general idea is to first predict an intermediate set of human-specified concepts $\mathbf{c}$ based on the input images and then use the concepts $\mathbf{c}$ to predict the target label $\mathbf{y}$. The objective is to address three desiderata: [36]

- **Interpretability**: Being able to note which concepts are important for the targets.
- **Predictability**: Being able to predict the targets from the concepts alone.
- **Intervenability**: Being able to replace predicted concept values with ground truth values to improve predictive performance.

These types of models are trained on datapoints $\{(\mathbf{x}^{(i)}, \mathbf{c}^{(i)}, \mathbf{y}^{(i)})\}_{i=1}^{n}$ where each input $\mathbf{x}$ (usually an image $\mathbf{x} \in \mathbb{R}^{H\times W\times C}$) is annotated with the set of concepts $\mathbf{c} \in \mathbb{R}^{k}$ with k concepts and the target $\mathbf{y} \in \mathbb{R}$. We can define the bottleneck model as $f(g(\mathbf{x}))$ where $g : \mathbb{R}^{H\times W\times C} \to \mathbb{R}^{k}$ maps an input $\mathbf{x}$ into the concept space and then $f : \mathbb{R}^{k} \to \mathbb{R}$ maps concepts into the target prediction. The bottleneck is performed when the target prediction $\hat{\mathbf{y}} = f(\hat{\mathbf{c}})$ relies only on the concept predictions $\hat{\mathbf{c}} = g(\mathbf{x})$ based on the inputs. An illustration of this architecture is shown in Figure **??**. For the training of these models, we have three different strategies:

- **Independent**: learn $\hat{f}$ and $\hat{g}$ independently. For $\hat{f}$, minimise the task loss $\mathcal{L}_Y$, which measures the discrepancy between predicted and true targets. Then, for $\hat{g}$, minimise the concept loss $\mathcal{L}_{C_j}$ for each concept j, which measures the discrepancy between predicted and real j-th concept.

$$\hat{g} = \arg\min_{g} \mathcal{L}_C(g(\mathbf{x}), \mathbf{c}), \quad \hat{f} = \arg\min_{f} \mathcal{L}_Y(f(\mathbf{c}), \mathbf{y}). \tag{94}$$

- **Sequential**: first learn $\hat{g}$ and then use the predictions to learn $\hat{f}$.

$$\hat{g} = \arg\min_{g} \mathcal{L}_C(g(\mathbf{x}), \mathbf{c}), \quad \hat{f} = \arg\min_{f} \mathcal{L}_Y(f(\hat{g}(\mathbf{x})), \mathbf{y}). \tag{95}$$

- **Joint**: minimises the weighted sum of both task and concept losses with the hyperparameter λ, which controls the tradeoff between concept vs. task loss.

$$\hat{g}, \hat{f} = \arg\min_{g,f} \left[\lambda \mathcal{L}_C(g(\mathbf{x}), \mathbf{c}) + \mathcal{L}_Y(f(g(\mathbf{x})), \mathbf{y})\right]. \tag{96}$$

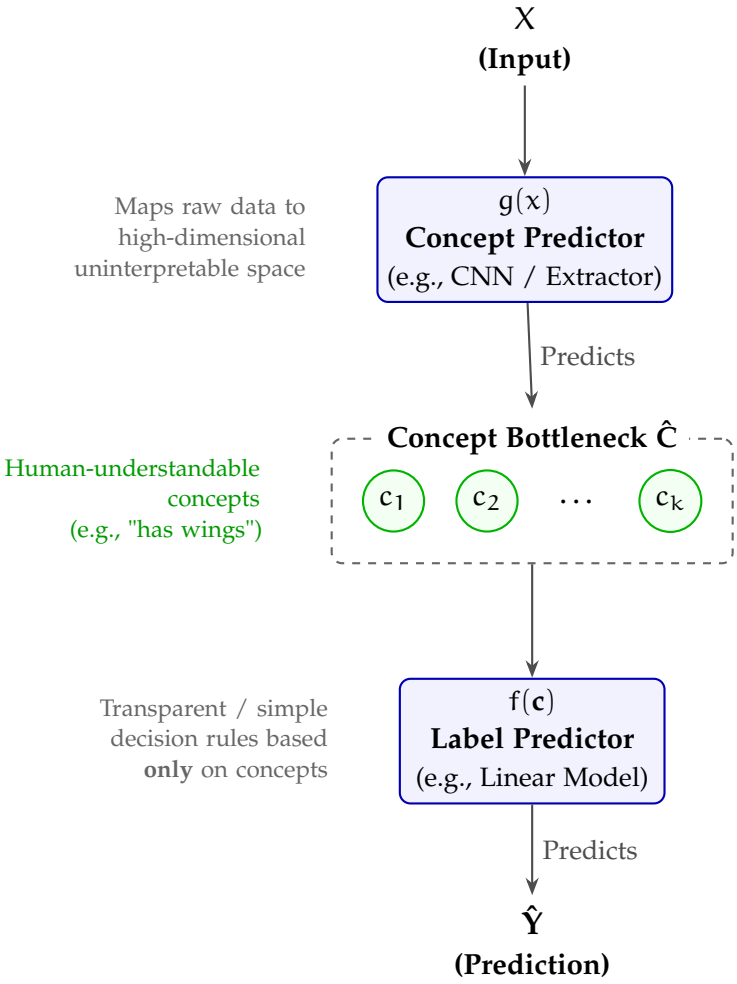


Figure 8: Architecture of a Concept Bottleneck Model (CBM). The model maps a raw inputs $\mathbf{x}$ to an interpretable intermediate concept space $\hat{\mathbf{c}}$ before masking the final prediction $\hat{\mathbf{y}}$.

One important property of these models is the ability to perform interventions by editing the concept predictions $\hat{c}$ and propagating these changes to the target prediction $\hat{y}$. This property also makes these models more interpretable in terms of high-level concepts since we can see the importance and effect of the inclusion or removal of some concepts for the final response. In the original paper, the concept intervention method is applied randomly without any specific criteria or using expert knowledge. In a recent paper [45], different intervention methods were evaluated to minimise the number of concepts to intervene on while maximising task accuracy. The method that achieves the best performance is the one where we first intervene on the concepts with the highest predictive uncertainty from the concept predictor g, i.e. it first intervenes on the concept with the maximum value of $1/|\hat{c}_i - 0.5|$.

8.1 INFORMATION LEAKAGE

For these types of models, it may often be unrealistic to assume that we will have access to concepts which fully capture the relationship between the inputs and targets. Several recent papers have studied the problem of information leakage in these types of models based on the previous assumption. This information leakage is usually observed in CBMs trained sequentially or jointly [35, 36], and it appears when the intermediate representations encode information that is not present in the concepts c, which contains information to predict the final target rather than only focusing on getting the concept correct. This leakage has a negative impact on the interpretability and intervenability of the model. A CBM model with information leakage may be unsafe for critical decision-making applications, such as clinical scenarios. [41] We can identify two main types of leakage:

- **Concept-task leakage**: additional information about the task is stored in the learnt concepts. If concept-task leakage is present, then the learned concept-task mutual information (MI) will be higher than ground-truth concept-task MI.
- **Interconcept leakage**: additional information about the other concepts is stored in each learned concept. Learned concepts are more predictive of the value of the other learned concepts, resulting in a learned interconcept MI being higher than the ground-truth.

Several measures of leakage have been proposed in the literature, but none of them has successfully integrated the information-theoretic nature of leakage in these types of models.

- **Concept Performance Metrics**: these are the metrics that indicate the quality of the concept learning, such as for example concept accuracy, precision, recall, F1 score and AUC. These metrics are not sensitive to leakage. Two models can have similar concept performance but have different degrees of information leakage. Concept AUC has more sensitivity in this problem since it contains more information about the concept distribution.
- **Oracle Impurity Score (OIS)**: this metric was defined in [9] to capture leakage. The idea is to train $k(k-1)$ additional simple MLP $\psi_{i,j}$, whose objective is to predict the value of the ground-truth concept j from the learnt representation of concept i. This will create the Purity Matrix, defined as $\pi(\hat{c}, c) = \mathrm{AUC}(\psi_{i,j}(\hat{c}_i), c_j)$. The (i,j)-th entry contains the AUC when predicting the ground truth value of concept j given the i-th concept representation. The OIS metric is the average difference

between the predictivities of learnt and ground truth concepts over pairs of concepts, defined as

$$\mathrm{OIS}(\hat{c}, c) = \frac{2}{k}\|\pi(\hat{c}, c) - \pi(c, c)\|_F$$

, where the Frobenius norm ensures $0 \leqslant \mathrm{OIS} \leqslant 1$. The value of 1 represents a complete misalignment; the i-th concept can predict all other concepts except its corresponding one. A value of 0 represents a perfect alignment; the i-th concept does not encode any unnecessary information for predicting concept i. A limitation is the stochasticity of the metric, which involves training neural networks, which is very variable when we evaluate the model several times at test time.

- **Niche Impurity Score (NIS)**: this metric was also defined in [9] to quantify the amount of shared information across concept representations. It is defined to capture the additional information about a concept j in a set of concepts that does not contain concept j. This metric appears to be anticorrelated with intervention performance, which produces doubts among the scientific community about its utility.

- **Intervention**: the intervention of the predicted concepts with the ground-truth values can be used to evaluate the interpretability and the leakage. A task accuracy that decreases after intervention indicates that the task head expects extra information that is not present in the ground-truth concepts. Models with higher leakage have worse intervention performance. This metric is unable to distinguish between interconcept and concept-task leakage; also, when we have a large number of concepts, the computation is very expensive.

Then, based on the limitations and drawbacks of the previous metrics, the authors of [41] presented two new metrics based on an information-theoretic perspective.

- **Concept-task Leakage Score** (CTL): for a concept i, the metric is defined as the difference between predicted and ground-truth mutual information between concept i and the task label, which quantifies the additional information that the concept i encodes about the task label with respect to the ground-truth.

$$\mathrm{CTL}_i(\hat{c}_i, c_i, y) = \max\left(0, \frac{I(\hat{c}_i, y)}{H(y)} - \frac{I(c_i, y)}{H(y)}\right), \tag{97}$$

where $H(y)$ is the entropy of the task labels and $I(c_i, y)$ is the mutual information between the ground-truth concepts and task labels. The general metric is the average between the k concepts.

- **Interconcept Leakage Score** (ICL): this is a similar metric that defines the pairwise interconcept leakage between concepts i and j, which is the additional predictivity of the learnt concept i for concept j with respect to ground-truth.

$$\mathrm{ICL}_{ij}(\hat{c}, c) = \max\left(0, \frac{I(\hat{c}_i, \hat{c}_j)}{\sqrt{H(\hat{c}_i)H(\hat{c}_j)}} - \frac{I(c_i, c_j)}{\sqrt{H(c_i)H(c_j)}}\right). \quad (98)$$

  We can define the per concept and average interconcept score by taking the average.

Calculating these metrics requires an efficient estimation of entropy and mutual information, which the authors achieved using the Kraskov-Stögbauer-Grassberger (KSG) estimator based on k-nearest neighbour statistics. Experimental results demonstrate that these metrics accurately detect varying degrees of leakage with minimal uncertainty, yielding scores that strongly correlate with intervention performance. Leveraging the robustness of these metrics, further analysis was conducted on the common causes of leakage in Concept Bottleneck Models (CBMs). First, models trained with low values of λ during joint training (Equation **??**) exhibit poor intervention performance, as task accuracy relies heavily on high leakage rather than accurate concept learning. While higher values of λ do not guarantee zero leakage and often degrade task performance, an optimal hyperparameter range for each model can be identified using CTL and ICL scores. Furthermore, if the representation chosen for the learned concepts is more expressive than the annotated concepts, the model exploits this excess capacity to store extraneous information. Another significant issue arises from incomplete concept sets; if the predefined concepts are insufficient, the model compensates by encoding missing variables into the learned concepts to boost predictive accuracy. Finally, if the classification head is not flexible enough to capture dependencies between concepts, the model inherently forces these dependencies into the concept representations themselves.

8.2 RECENT CBMS

A new framework for CBMs called Concept Embedding Models (CEMs) [54] was presented with the idea to overcome the accuracy vs interpretability trade-off found in concept incomplete settings. The general idea is to represent each concept as a supervised vector, where these high-dimensional embeddings allow for extra supervised learning capacity.
A model $\psi(\mathbf{x})$ learns a latent representation $\mathbf{h} \in \mathbb{R}^{n_{\text{hidden}}}$ which is the input of the CEM layer. The latent $\mathbf{h}$ is introduced into two concept-specific fully connected layers, which learn two concept embeddings in $\mathbb{R}^m$, $\hat{\mathbf{c}}_i^+ = \phi_i^+(\mathbf{h}) = a(\mathbf{W}_i^+\mathbf{h} + \mathbf{b}_i^+)$ and $\hat{\mathbf{c}}_i^- = \phi_i^-(\mathbf{h}) = a(\mathbf{W}_i^-\mathbf{h} +$

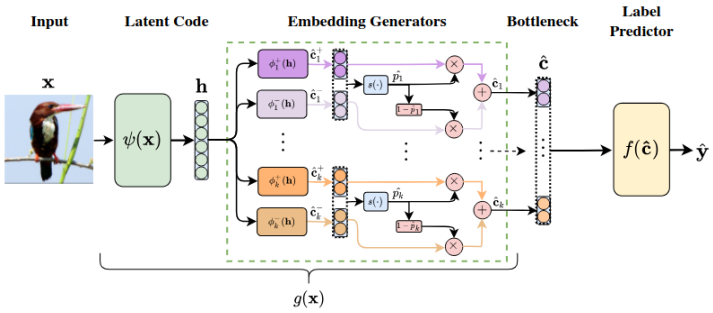


Figure 9: Illustration of the Concept Embedding Model framework.

$\mathbf{b}_i^-$). Then, we train a learnable scoring function trained to predict the probability of the concept c_i being active, $\hat{p}_i = s([\hat{\mathbf{c}}_i^+, \hat{\mathbf{c}}_i^-]^T) = \sigma(\mathbf{W}_s[\hat{\mathbf{c}}_i^+, \hat{\mathbf{c}}_i^-]^T + \mathbf{b}_s)$. In this case, the parameters $\mathbf{W}_s$ and $\mathbf{b}_s$ are shared across all concepts. The final concept embedding $\hat{\mathbf{c}}_i$ is defined as

$$\hat{\mathbf{c}}_i = \hat{p}_i \hat{\mathbf{c}}_i^+ + (1 - \hat{p}_i)\hat{\mathbf{c}}_i^-. \tag{99}$$

In Figure **??** we can visualize the general framework of CEMs. For the intervention, we only have to modify the concept probability with the ground-truth or the desired ones to build the intervened concept embedding. For the training of this model, they used the joint training strategy, and also, with a small probability, they calculate the concept embedding with the true concept probability.
Although the original paper of the CEMs claims an improvement in task accuracy, interpretability and steerability, other articles like [41] have demonstrated that this accuracy often comes at the cost of severe information leakage. This structural vulnerability undermines the model's actual interpretability, effectively breaking the core promise of the concept bottleneck. Information leakage in Concept Embedding Models (CEMs) primarily occurs because the high dimensional concept embeddings are over-expressive, allowing them to bypass the intended semantics and encode task-relevant shortcuts directly from the input. Specifically, the three major forms of leakage that were explained in the previous section are present in CEMs:

- **Concept-task Leakage**: this excess capacity allows the embeddings to capture additional task-predictive signals that are entirely unrelated to the underlying human-interpretable concept. Consequently, the model attains high predictive performance not because the concepts themselves are accurate, but because the embeddings act as covert channels for the task label.

- **Interconcept Leakage**: Instead of learning disentangled and independent features, the embeddings cluster based on the ground-truth values of neighboring concepts. Crucially, attempting to reduce this issue by increasing concept supervision in CEMs typically backfires, forcing even more interconcept leakage into the vector space.

- **Alignment Leakage**: because of the random interventions during training, the model learns to generate embeddings whose predictive power artificially depends on their alignment with the ground-truth concept state.

Taking inspiration of CEM and combining it with energy-based models, the method called Energy-based Concept Bottleneck Models (ECBMs) was proposed in [51]. This framework define a set of neural networks to define the joint energy of the input $\mathbf{x}$, concept $\mathbf{c}$ and class label $\mathbf{y}$. We consider a supervised classification setting with N samples, K concepts and M different classes, $\mathcal{D} = \{x^{(j)}, c^{(j)}, y^{(j)}\}_{j=1}^{N}$ where the j-th data points contains the input $x^{(j)} \in \mathcal{X}$, the label $y^{(j)} \in \mathcal{Y} \subset \{0,1\}^M$ and the concepts $c^{(j)} \in \mathcal{C} = \{0,1\}^K$. We denote as $\mathbf{y}_m$ the M-dimensional one-hot vector with the m-th dimension set to 1. Then, we denote $c_k^{(j)}$ denotes the k-th dimension of the concept vector $c^{(j)}$ and $[c_i^{(j)}]_{i\neq j}$ as $c_{-k}^{(j)}$ for brevity. We also have a pretrained neural network $F : \mathcal{X} \to \mathcal{Z}$ to extract the features $\mathbf{z}$ from the input $\mathbf{x}$.
This model consists of three energy networks with different objectives:

- **Class Energy Network** $E_\theta^{class}(\mathbf{x}, \mathbf{y})$. Each class m is associated with a trainable class embedding denoted as $\mathbf{u}_m$. The idea is to feed into a neural network $G_{zu}(\mathbf{z}, \mathbf{u})$ the features $\mathbf{z}$ from $\mathbf{x}$ and the embedding $\mathbf{u}$ associated with the class label $\mathbf{y}$. Formally, we have $E_\theta^{class}(\mathbf{x}, \mathbf{y}) = G_{zu}(\mathbf{z}, \mathbf{u})$, which uses the negative log-likelihood as the loss function defined as

$$\mathcal{L}_{class}(\mathbf{x}, \mathbf{y}) = -\log p_\theta(\mathbf{y}|\mathbf{x}) = E_\theta^{class}(\mathbf{x}, \mathbf{y}) + \log\left(\sum_{m=1}^{M} e^{-E_\theta^{class}(\mathbf{x}, \mathbf{y}_m)}\right). \tag{100}$$

- **Concept Energy Network** $E_\theta^{concept}(\mathbf{x}, \mathbf{c})$. This energy network consists of K sub-networks $E_\theta^{concept}(\mathbf{x}, c_k)$, which measure the compatibility of the input $\mathbf{x}$ and the k-th concept. Each concept k is associated with a positive embedding v_k^+ and a negative embedding v_k^-. The concept embedding v_k is defined as a weighted sum of positive and negative embeddings, weighted by the concept probability c_k. Formally, we define the concept energy network using a neural network $E_\theta^{concept}(\mathbf{x}, c_k) = G_{zv}(\mathbf{z}, v_k)$, where the loss function is defined as

$$\mathcal{L}_{concept}(\mathbf{x}, \mathbf{c}) = \sum_{k=1}^{K} \mathcal{L}_{concept}^{(k)}(\mathbf{x}, c_k), \tag{101}$$

  where the loss for each k-th sub-network is defined as

$$\mathcal{L}_{concept}^{(k)}(\mathbf{x}, c_k) = E_\theta^{concept}(\mathbf{x}, c_k) + \log\left(\sum_{c_k \in \{0,1\}} e^{-E_\theta^{concept}(\mathbf{x}, c_k)}\right).$$

(102)

- **Global Energy Network** $E_\theta^{global}(\mathbf{c}, \mathbf{y})$. This energy network learns the interactions among different concepts and between all concepts and the class label. We introduce into the neural network the $\mathbf{y}$ associated class label embedding $\mathbf{u}$ with the concatenation of the K concept embeddings, $[v_k]_{k=1}^K$. We formally have $E_\theta^{global}(\mathbf{c}, \mathbf{y}) = G_{vu}([v_k]_{k=1}^K, \mathbf{u})$, where the loss function is defined as

$$\mathcal{L}_{gobal}(\mathbf{c}, \mathbf{y}) = E_\theta^{global}(\mathbf{c}, \mathbf{y}) + \log\left(\sum_{m=1, \mathbf{c}' \in \mathcal{C}} e^{-E_\theta^{global}(\mathbf{c}', \mathbf{y}_m)}\right), \tag{103}$$

  where $\mathbf{c}'$ enumerates all concept combinations in the space $\mathcal{C}$.

8.3 CONCEPT BOTTLENECK GENERATIVE MODELS

All the previous models presented in this document are focused on classification or regression problems, which are the original implementations of concept bottleneck models. The next step is to combine this idea with generative models to develop a method for interpreting and intervening in them, called Concept Bottleneck Generative Models (CBGMs)[17]. This method allows us to steer the generative model, since we can change the concept of the image while preserving the image quality, and understand the model by identifying the important concepts that are responsible for the output.
The general idea is to insert a concept bottleneck (CB) layer into a generative model, which is divided in two parts: the pre-concept bottleneck part and the post-concept bottleneck part. The partition depends on the generative model we use (GANs, VAEs, Diffusion). The authors decided to use the Concept Embedding framework [54] for the concept bottleneck layer and also argue that it is unrealistic to expect that the pre-defined human concepts are complete, so they also added a set of unknown concepts that might also be required for generation.
Each concept is represented with two embeddings: $w_i^+, w_i^- \in \mathbb{R}^m$ representing the active and inactive concept states, respectively. The output of the pre-concept bottleneck part of the model, the embedding h, is fed into several context networks ϕ that map h to the two embeddings per concept. Then, a network Ψ is trained to predict the probability of concept c_i from the joint embedding space, $\hat{c}_i = \Psi_i([w_i^+, w_i^-]^T) \in [0, 1]$. The final context embedding w_i is defined as a weighted mixture of the two embeddings as $w_i = (\hat{c}_i w_i^+ + (1 - \hat{c}_i) w_i^-)$. All the k concept embeddings are concatenated to obtain the final vector that is fed to the post-concept network. For the intervention, we simply have to modify the concept probability $\hat{c}_i$ to $\bar{c}_i$ in

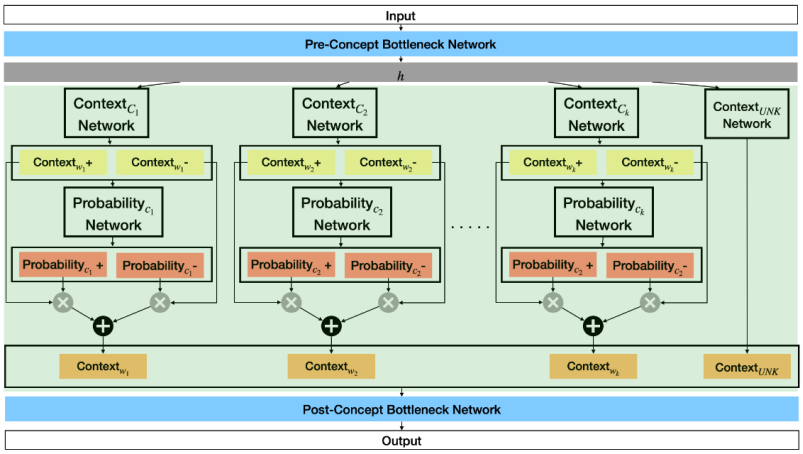


Figure 10: Architecture of the concept layer in the Concept Bottleneck Generative Model (CBGM) for a generic generative model.

the weighted mixture of w_i. A visualization of the concept layer is shown in Figure **??**. The loss function is defined as

$$\mathcal{L} = \mathcal{L}_{task} + \alpha\mathcal{L}_{con} + \beta\mathcal{L}_{orth},$$

where α and β are hyperparameters that control the importance of the concept and orthogonality loss, respectively. As we can see, the loss consists of three parts:

- **Task Loss**: the base objective function of the generative model family.
- **Concept Loss**: is the binary cross-entropy loss between the predicted and real concept probabilities.
- **Concept Orthogonality Loss**: encourage the unknown concepts to be orthogonal to the outputs of each concept network using an orthogonality constraint defined as:

$$\mathcal{L}_{orth} = \sum_{f \in B} \frac{\sum_{i=1}^{i=k} |\langle w_i, w_{k+1} \rangle|}{\sum_{i=1}^{i=k} 1},$$

  where $\langle \cdot, \cdot \rangle$ is the cosine similarity and j denotes each sample in the mini-batch of size B.

One of the main limitation of the previous method is that the generative model needs to be trained from scratch with the new concept bottleneck layer and using a dataset with ground-truth concept annotations. To build a more efficient and scalable method, we can use post-hoc generative concept bottleneck models [26], which can transform any pre-trained generative model into a interpretable one with concept bottleneck layer.
Based on the setup of the previous method, where we divided the

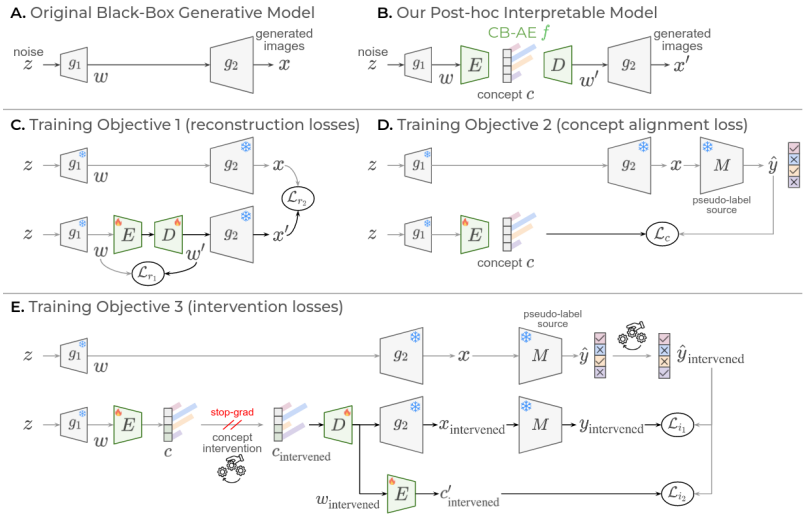


Figure 11: Architecture of the Post-hoc Concept Bottleneck Generative Model where the different loss functions are illustrated.

generative model in two parts ($g = g_2 \circ g_1$, $\mathbf{z}$ initial noise, $\mathbf{x}$ generated sample where $\mathbf{x} = g_2(g_1(\mathbf{z}))$), the idea is to include a concept bottleneck autoencoder. The latent space of the autoencoder is the concept prediction $\mathbf{c} = E(g_1(\mathbf{z}))$. This concept prediction is defined using the concept embedding framework and also includes a set of unknown concepts to complement the human-defined concepts. Then, the decoder D reconstructs the features from the fist part of the generative model (g_1) based on the concept prediction $\mathbf{c}$, $\mathbf{w}' = D(\mathbf{c})$, where $\mathbf{w} = g_1(\mathbf{z})$ are the original features. Now we can generate samples from the original image features $\mathbf{w}$ as well as from the reconstructed features $\mathbf{w}'$, defined as $\mathbf{x}' = g_2(\mathbf{w}') = g_2(D \circ E(\mathbf{w}))$. For the training of this model, where the generative model is keep frozen, the authors defined three different objective functions based on the different desired goals:

- **Reconstruction Losses**: we sample $\mathbf{w}$ from the first part of the generative model and use it to generate the original image $\mathbf{x}$. The, we introduce $\mathbf{w}$ in the autoencoder to get the reconstructed features $\mathbf{w}'$, which is used to generate the reconstructed image $\mathbf{x}'$. The reconstruction losses are defined as the distance between the original and reconstructed features and images, respectively. The definition is the following:

$$\min_{E,D} \left[\mathcal{L}_{r_1}(\mathbf{w}, \mathbf{w}') + \mathcal{L}_{r_2}(\mathbf{x}, \mathbf{x}') \right]. \tag{104}$$

- **Concept Alignment Loss**: first we have to obtain the pseudo-labels $\hat{\mathbf{y}} = M(\mathbf{x})$ from a generated image $\mathbf{x}$. This model M can be a supervised model or a zero-shot model, such as CLIP. Then, we can obtain the concept prediction $\mathbf{c} = E(g_1(\mathbf{z}))$ and use it to generate the image $\mathbf{x}'$. We can also obtain the pseudo-labels

$\hat{\mathbf{y}}' = M(\mathbf{x}')$ from the generated image. The concept alignment loss is defined as the distance between the pseudo-labels of the original and the predicted concepts. The definition is the following:

$$\min_{E} \mathcal{L}_c(\hat{\mathbf{y}}, \mathbf{c}). \tag{105}$$

- **Intervention Losses**: they added a third loss to encourage the steerability of the model, which is an important property of CBMs. The idea is to encourage the decoder D to produce realistic $\mathbf{w}'$ when the concepts are modified. First, we have to select a concept i to intervene by swaaping the logits in the latent concept space to obtain the $\mathbf{c}_{int}$. With the intervened concepts, we can reconstruct the intervened features $\mathbf{w}_{int}$ and the intervened image $\mathbf{x}_{int}$, and using the pseudo-labeler, we can get the concept prediction $\mathbf{y}_{int} = M(\mathbf{x}_{int})$. The final step is to intervene the pseudo-label of the original image by changing only the concept i and get $\hat{\mathbf{y}}_{int}$. The intervention loss is defined as the distance between the intervened pseudo-labels, which encourages the model to generate images that reflect the concept changes. The definition is the following:

  $$\min_{E,D} \mathcal{L}_{i_1}(\hat{\mathbf{y}}_{int}, \mathbf{y}_{int}). \tag{106}$$

  Also, to improve consistency, they pass the intervened features $\mathbf{w}_{int}$ through the encoder to get the concept predction $\mathbf{c}'_{int}$ and apply cross-entropy loss defined as:

  $$\min_{E,D} \mathcal{L}_{i_2}(\hat{\mathbf{y}}_{int}, \mathbf{c}'_{int}). \tag{107}$$

The next step is the method called Concept Bottlenecks on Energy Landscapes (CoBEla) [22], which presents a decoder-free and energy-based framework that use a pre-trained frozen generative model. Assume we have the same setup as before, with a generative model divided in two parts and a pseudo-labeler. They perturb the intermediate features using the DDPM cosine scheduler to get $\mathbf{w}_t$. For each concept k, they define an energy network E_θ that takes as input the perturbed features $\mathbf{w}_t$ and the concept c_k, which is a learnable concept embedding. The encoder outputs two logits and it is defined as two conditional residual blocks with FiLM conditioning [42]. The concept score is defined as the positive-class probability, which leads to the definition of the interpretable bottleneck score vector $\mathbf{s} = [s_1, s_2, \ldots, s_K]$. Then, the logits are converted into scalars to represent the energy for each concept, where the aggregation of per-concept energies is defined as $\mathcal{E}_\theta(\mathbf{w}_t)$, which is used in the score matching loss (defined in Section – and Equation –) as the noise prediction of the energy-based model. As a complementary loss, we have

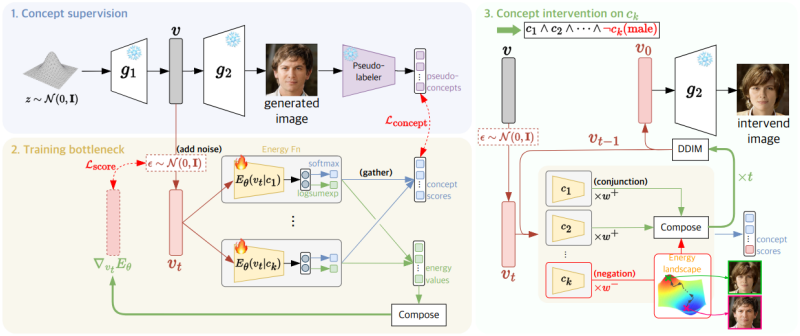


Figure 12: Architecture of the CoBEla Model where the training and guided generation processes are illustrated.

the concept loss that compare the per-concept logits with the pseudo-labels. The total objective is defined as

$$\mathcal{L}(\theta) = \lambda_1 \mathcal{L}_{score} + \lambda_2 \mathcal{L}_{concept}. \tag{108}$$

The visualization of the framework is shown in Figure **??**. In test time, we assign an intervention weight to each concept that yields to the final weighted energy defined as:

$$\mathcal{E}_\theta(\mathbf{w}_t) = \sum_{k=1}^{K} \lambda_k E_\theta(\mathbf{w}_t, c_k). \tag{109}$$

This model allows for the conjunction and negation of concepts, which is a very interesting property that allows for more complex interventions. They used the DDIM sampling method to obtain the final generated intermediate features using the weighted energy, which is then fed to the second part of the generative model to get the final generated image.

Part II

METHODS

9 PROPOSAL

This thesis aims to analyze and improve current methods for interpretability, compositionality, and disentangled representation learning in state-of-the-art generative models. The ultimate goal is the generation of reliable, controllable, and safe medical images. Grounded in theoretical formulation, model development, and empirical evaluation within the medical domain, the execution of this project is structured into the following core components:

9.1 THEORETICAL AND EXPERIMENTAL MODEL DEVELOPMENT

The foundational phase of the project focuses on comprehensively understanding, implementing, and modifying baseline generative models. This ensures high-quality image fidelity while establishing robust frameworks to evaluate interpretability and controllability.

- **Mathematical Foundation**: Rigorously study the mathematical principles of diffusion and flow-based models, including score-based generative modeling, Ordinary and Stochastic Differential Equation (ODE/SDE) based sampling, and path-based formulations such as Flow Matching.
- **Disentanglement/Compositionality Theory**: Synthesize the theoretical frameworks underlying disentangled and compositional representation learning through a critical analysis of state-of-the-art methodologies and standard evaluation metrics.
- **Implementation**: Implement, reproduce, and benchmark baseline generative models using Python and PyTorch.
- **Controllability and Compositionality Design**: Propose and integrate architectural and training modifications to enhance control mechanisms. This includes developing techniques for spatial and semantic conditioning, guidance analysis, and disentangled representations (e.g., slot-based generation) to allow the independent manipulation of specific pathological or anatomical features.

9.2 DATA ACQUISITION AND PREPARATION

High-quality, standardized data is critical for training robust generative models and evaluating their compositional capabilities.

- **Dataset Sourcing**: Utilize established, publicly available general-purpose datasets for initial baselines, transitioning to specialized medical imaging datasets relevant to the targeted anatomies and pathologies.
- **Preprocessing Pipelines**: Apply comprehensive data preprocessing pipelines, including intensity normalization, spatial resizing, and data augmentation. Furthermore, implement latent-space encoding using pre-trained Autoencoders or Variational Autoencoders (VAEs) to compress high-dimensional medical images for efficient generative modeling, as in Latent Diffusion Models.

9.3 MODEL TRAINING AND EVALUATION

The training phase leverages advanced distributed computing strategies to handle the vast computational requirements of modern generative frameworks.

- **Distributed Training**: Train models on high-performance computing infrastructure using large-scale distributed training paradigms (such as Distributed Data Parallel (DDP)) to accelerate experimentation and scaling.
- **Quantitative Metrics**: Evaluate the fidelity, diversity, and accuracy of the generated data using standard generative metrics, including Fréchet Inception Distance (FID), Kernel Inception Distance (KID), Precision, and Recall, alongside specific metrics for disentanglement representation learning and compositional evaluation.

10 DATA

In this section, I want to introduce the different datasets that I have used during this part of my research and the possible future datasets that I plan to use in the remaining part of my thesis. In this initial part, where I have focused on the implementation of some methods found in the literature along with some small experiments to have an intuition about the field, I have used a collection of small and well-known in the literature datasets. These datasets are the following:

- **MNIST** [28]: a dataset of handwritten digits with 60,000 training samples and 10,000 test samples, where each sample is a 28x28 grayscale image of a digit from 0 to 9.
- **Shapes2D** [33]: a dataset of 2D shapes with 737,280 samples, where each sample is a 64x64 RGB image of a shape (square, ellipse or heart) with different colours and sizes. The factors of variation of each data sample are known
- **Shapes3D** [33]: a dataset of 3D shapes with 480,000 samples, where each sample is a 64x64 RGB image of a 3D shape (cube, sphere or cylinder) with different colors and sizes.
- **dSprites** [37]: a dataset of 2D sprites with 737,280 samples, where each sample is a 64x64 grayscale image of a sprite with different shapes, colours, and positions. We also know the 6 ground truth independent factors of variation, which are colour, shape, scale, rotation, x and y positions of a sprite.
- **CelebA** [32]: a dataset of celebrity faces with 202,599 samples, where each sample is a 178x218 RGB image of a celebrity face with different attributes (gender, age, hair color, etc.).
- **CelebA-HQ** [19]: a dataset of high-quality celebrity faces with 30,000 samples, where each sample is a 1024x1024 RGB image of a celebrity face with different attributes (gender, age, hair color, etc.).

Several articles related to disentanglement representation learning use these datasets because they include the ground-truth factors of variation. Because of that, it is important to also use these datasets to evaluate our experiments to have a fair comparison with other methods in the literature.

The main objective of this thesis is to apply disentanglement and

compositionality methods in the generation of medical domains. To achieve this, we have to work with medical image datasets, where the idea is to use them once the method is validated in some of the previous datasets. Some medical datasets that I found interesting for my thesis are

- **NIH-Chest-X-ray-Dataset**. Curated by the National Institutes of Health, this dataset comprises 112,120 frontal-view X-ray images from 30,805 unique patients. It includes text-mined disease labels corresponding to 14 different thoracic pathologies. The variety of overlapping conditions within the images makes it a highly suitable benchmark for evaluating how well disentangled representations can isolate and control distinct clinical factors.

- **CheXpert: Chest X-rays** [16]. This dataset contains 224,316 chest radiographs of 65,240 patients who underwent a radiographic examination from Stanford Health Care between October 2002 and July 2017. For each image, the dataset contains labels for the presence of 14 common chest radiographic observations, which are extracted from free-text radiology reports and capture uncertainties present in the reports by using an uncertainty label.

- **MIMIC-CXR Database**. This is a large, publicly available dataset consisting of 377,110 chest radiographs corresponding to 227,827 imaging studies. Sourced from the Beth Israel Deaconess Medical Center, it uniquely pairs high-resolution medical images with their associated free-text radiology reports. The integration of complex visual data and rich clinical narrative provides an excellent foundation for exploring multi-modal compositionality.

Part III

EXPERIMENTS AND RESULTS

11

UNCONDITIONAL FLOW MATCHING MODEL FOR X-RAY IMAGES

As discussed in the literature review, contemporary techniques frequently modify existing pre-trained generative models, often by integrating supplementary components or introducing latent space disentanglement, to enhance performance, interpretability, or compositionality without compromising computational efficiency. However, these foundational models are predominantly trained on vast, general-domain datasets rather than the specialized medical data central to this thesis. To bridge this domain gap, an unconditional flow matching model was trained directly on a medical dataset, thereby establishing a tailored, pre-trained baseline for subsequent experiments. Rather than aiming for SOTA performance, the primary objective of this phase is to construct a foundational baseline to facilitate downstream investigations into disentanglement and compositionality within the medical domain.

This experiment utilizes the NIH Chest X-ray dataset (detailed in Chapter **??**) in combination with the Representation Autoencoder (RAE) framework [55]. In this approach, image latent representations are extracted using a pre-trained visual model—specifically, a DINOv2-Base architecture configured with 4 register tokens [7, 40]. The original RAE methodology involves training a decoder to form an autoencoder from a large-scale pre-trained vision model. This yields latent representations with enhanced semantic depth and superior reconstruction fidelity compared to the standard SD-VAE latents prevalent in state-of-the-art Stable Diffusion frameworks. The RAE authors subsequently trained a flow matching model on these latents using a Diffusion Transformer (DiT). While this achieved strong generative performance, it incurred significant computational costs due to the high-dimensional nature of the latents. To mitigate this computational bottleneck, they proposed a modified DiT architecture incorporating design principles from Decoupled Diffusion Transformer (DDT) models [49], which circumvents the necessity of scaling network width across the entire backbone. When evaluated on ImageNet, this modified model achieved SOTA performance in both Fréchet Inception Distance (FID) and Inception Score (IS) compared to alternative pixel and latent diffusion models.

Motivated by the high generation quality demonstrated in the RAE framework and the semantic utility of its latents, a comparable model

was trained specifically on the NIH Chest X-ray dataset. For this implementation, the standard DiT-XL architecture was selected over the modified DiT^{DH}-XL variant. This decision was deliberately made to maintain a distinct informational bottleneck, which is essential for facilitating subsequent experiments inspired by concept bottleneck generative models. The improved architecture features an input bypass directly to the final projection head, a structural element that complicates latent interpretability and reduces the precision of targeted latent interventions.
The comprehensive hyperparameter configuration and training details are provided in Table **??**. The model was trained over 450 epochs, requiring approximately 48 hours of compute time across four NVIDIA Hopper H100 GPUs.

Following the training phase, the generative performance of the model was evaluated using FID, IS, Precision, and Recall. This evaluation was conducted by generating 6,000 synthetic samples and comparing them against the pre-computed features of 10,000 real images from a randomly partitioned validation set. Because the model is unconditional, standard Classifier-Free Guidance (CFG) [13] was not applicable during sampling. Instead, AutoGuidance (AG) [20] was evaluated to determine if sample quality could be further enhanced. In the AG framework, the final prediction is formulated as a linear combination of the output of the final model and that of a lower-capacity auxiliary model (trained for a fraction of the total epochs).
The quantitative evaluation results are summarized in Table **??**, with a qualitative visualization of four generated samples provided in Figure **??**. The baseline model achieved an FID of 25.44 and an IS of 2.27 (improving marginally to 25.42 and 2.28, respectively, with AG). These metrics represent a solid baseline given the high structural complexity of the dataset and the unconditional nature of the generation process. Furthermore, the precision and recall scores indicate that the model successfully synthesizes a diverse distribution of images while maintaining an acceptable level of visual fidelity. Nevertheless, the FID score highlights an area for future optimization, suggesting that the statistical distribution of the generated samples could be further aligned with the underlying real data distribution.

Table 1: Hyperparameter configuration used for training.

Category	Hyperparameter	Value
Optimization	Optimizer	AdamW
	Learning rate	2×10^{-4}
	Weight decay	0.0
	Gradient clip norm	1.0
Schedule	Warmup epochs	40
	LR schedule	Linear
	Final learning rate	2×10^{-5}
Architecture	Input size	16
	Patch size	1
	In channels	768
	Hidden dimension	1152
	Depth	28
	Attention heads	16
	Dropout	0.1
Training	Batch size	128
	Mixed precision	`float32`
	EMA decay	0.995
	Training epochs	450
	Training steps	222

Table 2: Performance metrics of the DiT-XL trained on the NIH-Chest-X-ray dataset.

Model	Epochs	#Params	FID↓	IS↑	Precision↑	Recall↑
DiT-XL	450	838.68M	25.44	2.27	0.28	0.45
DiT-XL - AG	450	838.68M	25.42	2.28	0.29	0.45

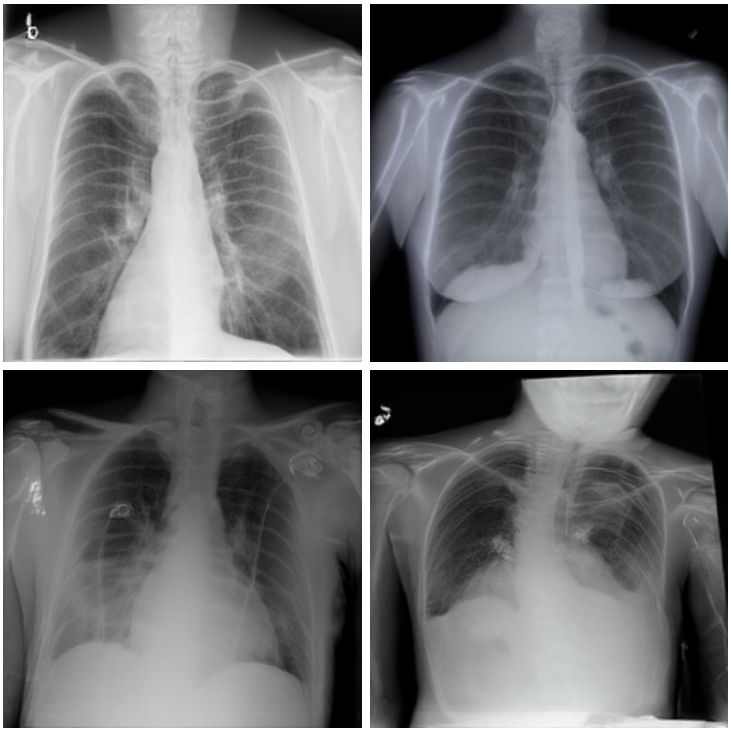


Figure 13: Example of 4 generated X-ray images using the trained DiT-XL model and AutoGuidance (AG).

12

SLOT ATTENTION WITH CONCEPT SUPERVISION IN CELEBA-HQ

Following the review of Slot Attention and other Object-Centric Learning (OCL) methods in Section **??**, natural connections to Concept Bottleneck Models (CBMs, Section **??**) emerge. Both paradigms inherently seek to decompose complex, high-dimensional visual inputs into a discrete set of interpretable variables prior to downstream reasoning. In both frameworks, the architecture forces the model to route its decision-making process through a constrained intermediate layer, whether comprising a set of permutation-invariant slots representing physical entities or a vector of semantic concepts aligned with human logic.

Despite this shared structural philosophy, fundamental differences exist in their objectives and learning signals. OCL methods, such as Slot Attention, are traditionally unsupervised, relying on spatial competition and reconstruction objectives to discover emergent perceptual groupings. Conversely, CBMs are explicitly supervised, requiring ground-truth labels to map visual features directly to predefined, high-level semantics. Furthermore, while traditional CBMs often struggle to spatially ground their concepts within specific image regions, Slot Attention excels at isolating localized areas, but without the guarantee that a discovered slot will align with a human-interpretable trait.

Recognizing the complementary strengths of these approaches, this section proposed a new architectural bridge. By applying explicit concept supervision to a Slot Attention module, this framework aims to spatially ground predefined facial attributes from the CelebA-HQ dataset directly into distinct visual slots. Drawing inspiration from the CODA architecture [38], this model also introduces several register tokens designed to capture residual information unaligned with the predefined semantic concepts.

Encoder. The encoder builds upon the foundation established in CODA, initially extracting DINOv2 features from the input image before applying Slot Attention to derive slot embeddings. However, aligning these supervised slots with a predefined set of concepts necessitates a fundamental shift in slot initialization. Unlike standard Slot Attention models, which initialize slots utilizing a single shared Gaussian distribution, the proposed architecture employs a heterogeneous initializa-

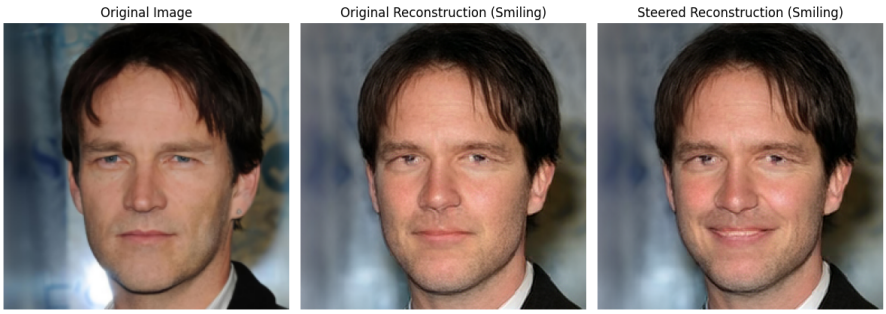


Figure 14: Example of reconstructed image with the Autoencoder with supervised Slot Attention and concept intervention for the slot Smiling.

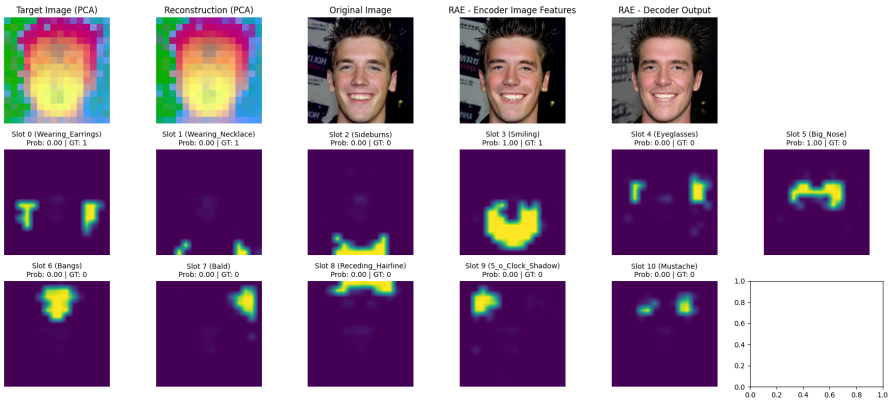
tion strategy via the reparameterization trick. Each supervised concept slot is assigned a unique, learnable prior defined by a mean parameter μ_c and a log-variance parameter $\log \sigma_c$. These concatenated initial slots, alongside the flattened spatial features, are subsequently fed into the Slot Attention module. Through iterative cross-attention, the slots compete for portions of the visual input, dynamically binding to distinct visual entities or global contexts. To enforce semantic alignment, the final concept slots are processed through a series of independent linear classifiers. Each classifier strictly corresponds to a single slot, outputting concept logits. This architectural constraint explicitly disentangles the latent space, forcing specific slots to encode designated semantic attributes.

Decoder. The decoder is designed to reconstruct the spatial feature map from the unordered, abstract slots. It leverages a Transformer-based cross-attention mechanism to map the abstract latent space back into the DINOv2 feature space. A critical design choice within the decoder is the explicit bifurcation of slot pathways. Rather than processing all slots uniformly, the decoder applies distinct Key (K) and Value (V) linear projections for the object slots versus the register slots:

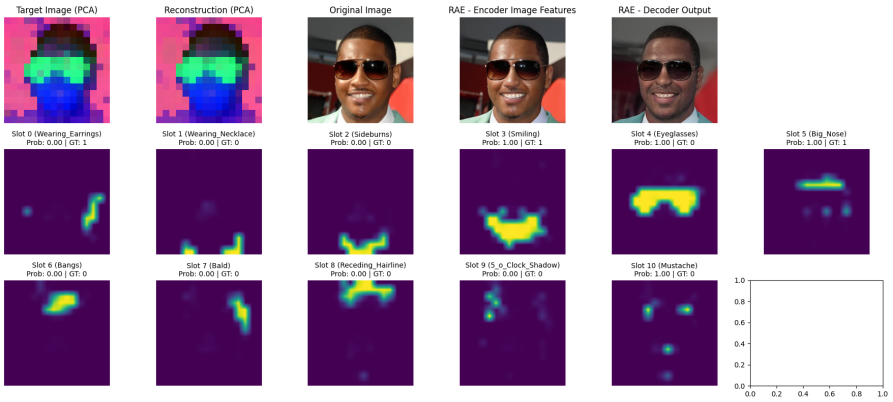
$$K_{obj}, V_{obj} = \text{Proj}_{obj}(S_{obj}) \tag{110}$$

$$K_{reg}, V_{reg} = \text{Proj}_{reg}(S_{reg}) \tag{111}$$

These separated pathways are subsequently concatenated. This bifurcation enables the network to process spatial object representations distinctly from the global scene statistics or background context captured by the four register tokens. The output yields the reconstructed DINOv2 features of the input image. Ultimately, this architecture forms an Autoencoder where the latent representation consists of the supervised slot embeddings. To generate the final reconstructed image, a pre-trained RAE decoder is utilized to map the reconstructed image features back into the pixel space.



(a)



(b)

Figure 15: Visualization of the PCA projection of the DINOv2 features, original and reconstructed images and attention maps for the set of supervised slots in two examples.

To evaluate the efficacy of the training process, Principal Component Analysis (PCA) projections of both the true and reconstructed DINOv2 features are visualized alongside the original images, the reconstructed images, and the slot-specific attention maps. Figure **??** presents two qualitative examples demonstrating that the attention maps successfully align with the corresponding visual pixels for the majority of the learned concepts. Notably, the Smiling concept is accurately localized in both examples, and the Eyeglasses concept in the second image yields a precise segmentation of the glasses.

A defining advantage of integrating CBM principles into this framework is the capacity for targeted concept intervention. The objective is to perturb a single slot within the latent space such that the decoded image exhibits variations exclusively in the pixels associated with that specific concept. The intervention protocol operates as follows: first, the learned weight vector is extracted from the slot's dedicated linear classifier. This vector serves as the normal to the decision

boundary, pointing in the direction of steepest ascent for the target attribute. To ensure the intervention magnitude remains consistent and independent of the classifier's original weight scale, this normal vector is L2-normalized into a unit direction vector. Subsequently, the normalized vector is scaled by a predefined steering strength and algebraically added to the selected slot representation. Through this precise vector addition, the intervention deliberately pushes the slot deeper into the positive half-space of the classifier to amplify the concept, or subtracts from it to suppress the feature. This mechanism provides granular control over the latent space prior to decoding. Figure **??** illustrates an intervention where the Smiling concept is artificially amplified. The resulting image exhibits targeted alterations strictly associated with the smile, preserving all other visual elements and confirming the precision of the proposed intervention methodology.

Part IV

RESEARCH PLAN

13

TIMELINE

In this chapter I want to update the timeline of my PhD based on the one presented in the initial Research Plan. The project can be organized in different stages, some of them are the same as the ones presented previously while others are removed or added. An updated Gantt chart showing the different stages and their expected duration is also included in Figure **??**.

- **Stage 1: Literature Review and Theoretical Foundations**. This stage involves an in-depth review of the existing literature on deep generative models (VAEs, Diffusion, and Flow Matching), disentangled representation learning, compositionality, concept bottleneck models, and generative AI for the medical domain. By studying the theoretical foundations of these topics, I aim to identify key challenges, formulate research questions, and discover gaps and opportunities for my own research. Additionally, this review will help me define the quantitative and qualitative metrics that will be used to evaluate the performance of my proposed methods.
  Originally, this stage was scheduled to be completed within the first six months of the PhD. While this timeframe is realistic for mastering the basic foundations of generative models, a comprehensive literature review is inherently a continuous process. Because the field of generative AI is evolving so rapidly, reviewing new papers and developments is an ongoing necessity throughout the PhD. I have already made significant progress in this area by reviewing a vast body of relevant literature and identifying the core challenges of existing methods. To better reflect this reality, my updated timeline extends the primary literature review phase through Q1 of my second year.

- **Stage 2: Preliminary experiments**. In this stage, the goal is to conduct preliminary experiments to explore the capabilities and limitations of existing models regarding disentanglement and compositionality. This process will help me establish baseline models and define metrics to evaluate the performance of my future proposed methods. Specifically, this includes reproducing key generative models, such as VAEs, Diffusion, and Flow Matching, and implementing various disentanglement and compositionality metrics.
  Initially, this stage was scheduled for completion between Q3 and Q4 of my first year. In practice, I began these preliminary experiments slightly later, in Q4. Furthermore, I realized that

preliminary experimentation is not a brief phase that can be concluded in a few months. Instead, it heavily overlaps with the literature review, as new methods and architectures are constantly emerging. Consequently, I have extended this stage through Q4 of my second year; I have found that actively implementing and experimenting with these models is an excellent way to deepen my understanding of the literature and identify ongoing challenges. I have already implemented several key models and metrics, and some of these initial results are presented in the experiments section.

- **Stage 3: Method Development and Architectural Innovation**. In this stage, the objective is to develop new methods and architectures for deep generative models that achieve better disentanglement, compositionality, and controllability. This will involve designing novel model architectures, training pipelines, and evaluation metrics to accurately assess performance. Specifically, I plan to implement modifications, extensions, and combinations of existing generative models with external frameworks to enhance their overall capabilities.
  Originally, this stage was scheduled for months 12 through 24 of the PhD. In the updated timeline, I have shifted this phase to run from Q3 of my second year through Q2 of my third year. This allows for a natural overlap with the preliminary experimentation stage, as the empirical insights gained there will directly inform the development of these new approaches. I have already begun drafting preliminary concepts and plan to continue refining and testing these ideas in the coming months.

- **Stage 4: Application to the Medical Domain**. In this stage, I will apply the developed methods to the medical domain, specifically targeting the analysis of medical images and addressing domain-specific challenges such as resolution, structure preservation, and clinical realism. To assess practical viability and gather expert feedback, I may establish collaborations with medical professionals or institutions. Additionally, ethical, legal, and data governance considerations will be carefully reviewed. During this phase, I will begin working with curated medical image datasets to evaluate the proposed methods, focusing on their ability to generate high-quality, clinically relevant images and provide insights into the underlying data.
  Originally, this stage was planned for months 24 through 30 of the PhD. In the updated timeline, I have shifted it to run from Q2 of my third year through Q1 of my fourth year. This adjustment allows for some overlap with the previous stage, as the specific challenges and requirements of the medical domain

will directly inform the ongoing development of the new methods and architectures.

- **Stage 5: Evaluation, Validation, and Writing**. In this final stage, I will conduct a comprehensive evaluation of the proposed methods, encompassing both quantitative metrics and qualitative assessments. I will validate these results through close collaboration with domain experts and prepare the findings for publication in peer-reviewed journals and conferences. To ensure reproducibility and contribute to the open-science community, I also plan to make the codebases and trained models publicly available. Additionally, I will write the final thesis document, synthesizing my individual research contributions into a cohesive narrative, and outline promising directions for future research. Finally, this phase will culminate in the preparation for my thesis defense.
  This stage is scheduled for the final year of the PhD, specifically running from Q2 through Q4 of my fourth year. Naturally, there will be some overlap with the preceding application stage, as the evaluation and clinical validation of the models will be an iterative, ongoing process throughout the final phases of development.

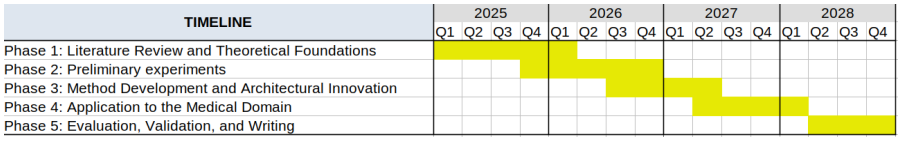


Figure 16: Updated timeline of the PhD project, showing the different stages and their expected duration.

14

DATA MANAGEMENT PLAN

This project relies entirely on pre-existing, curated, and de-identified datasets. No new primary data collection involving human subjects will be conducted. The data management strategy is structured in three progressive phases, transitioning from general-purpose computer vision datasets to restricted-access clinical datasets.

The research will utilize the following datasets, categorized by project phase:

- **Phase 1: Prototyping and Baseline Implementation**: Initial architectural development, debugging, and proof-of-concept experiments for disentangled generation and compositionallity analysiswill be conducted using well-known, publicly available datasets. These include MNIST, Shapes3D, and CelebA. See Chapter **??** for a detailed list and description of the datasets. These datasets are open-access and require no special data use agreements.
- **Phase 2: Medical Domain Adaptation**: To transition the generative models to the medical domain, publicly available medical imaging datasets will be utilized. While initial experiments will leverage established repositories like the NIH Chest X-ray and CheXpert datasets, this research is not restricted to radiography. The proposed methods are designed to be adaptable across diverse medical imaging modalities (e.g., MRI, CT, or histopathology). These datasets are fully de-identified and openly available for research purposes, providing a safe and ethical environment to test the models on complex anatomical structures and a wide variety of pathologies. A detailed description of these datasets can be found in Chapter **??**.
- **Phase 3: Advanced Clinical Application**: For the final evaluation of compositional generation and clinical realism, the project will utilize the MIMIC-CXR dataset. This is a large, restricted-access database of chest radiographs. Access to the MIMIC-CXR dataset has been formally granted via PhysioNet. I have completed the required credentialing (e.g., CITI training on Data or Specimens Only Research) and signed the Data Use Agreement (DUA). No attempts will be made to re-identify any patients. The original restricted data will not be copied to unauthorized personal devices, and access will be limited solely to credentialed researchers involved directly in this project.

All datasets will be securely stored at the Barcelona Supercomputing Center (BSC - MareNostrum 5) and on password-protected, encrypted local workstations used for prototyping. Throughout the project, the generative models will produce purely synthetic medical data (e.g., artificial radiographs and corresponding segmentation masks or labels). This synthetic data contains no real patient information and poses no privacy risks. Derived data, such as model weights, evaluation metrics, and latent embeddings, will also be generated and stored securely on the BSC infrastructure.

15

TRAINING PLAN

I will engage in a comprehensive training plan designed to strengthen my theoretical foundation, enhance my technical proficiency, and foster integration within the broader research community. As training activities are completed, they will be properly documented and recorded in the Doctoral Student Activity Document (DAD). This record will serve as a living document, continually updated throughout my doctoral studies.

Throughout my PhD, I will participate in transversal training provided by the Doctoral School, logging these activities in DRAC to maintain my DAD. Specifically, I plan to undertake coursework related to open science and citizen science beginning in my second year, continuing into the third and fourth years. Additionally, I will cultivate essential soft skills—such as communication, leadership, and project management—through workshops, seminars, and courses offered by both the Doctoral School and the Barcelona Supercomputing Center (BSC). I will also actively attend conferences and workshops in the fields of deep generative models and medical imaging to network with peers and exchange novel ideas.

My specialized training is already underway. In the summer of my first year, I attended the "MLx Representation Learning & Generative AI" summer school at the University of Oxford. This program provided a comprehensive overview of state-of-the-art techniques, led by speakers at the forefront of the field. For the summer of my second year, I plan to attend the Probabilistic AI Summer School in Lithuania, which focuses on variational inference, diffusion models, and related topics. In the subsequent years of my PhD, I will continue to participate in relevant summer schools and conferences to stay abreast of emerging trends and expand my professional network.

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